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SECTION 1

TITLE:

ENTERPRISE SYSTEM DEVELOPMENT AND MANAGEMENT (ESDM) PROJECT

LECTURER: DR. ARYATI BINTI BAKRI

NAME OF STUDENTS	MATRIC NO.
FARRA NURZAHIN BINTI ZAHARIL ANUAR	A23CS0079

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Universiti Tun Hussein Onn Malaysia (UTHM) is a public higher education institution that manages diverse academic and administrative activities to support students, academic staff, and university management. These functions are supported by several core operational information systems that collectively form the university's enterprise systems environment. As the scale and complexity of university operations increase, the need for structured coordination between business processes, information systems, and supporting technologies becomes more significant.

Currently, UTHM utilizes a basic form of enterprise architecture to support its information systems landscape. While this provides a general structure for system usage and management, it does not yet represent a comprehensive or formally defined enterprise architecture that fully aligns business needs with application, data, and technology layers. As a result, there is an opportunity to further develop and formalize the enterprise architecture to improve system integration, consistency, and long-term planning.

This project involves the development of a proposed enterprise architecture for UTHM that covers four core operational systems at the enterprise level. Each system is analyzed by a different project group to provide detailed architectural insights. The focus of this project is on Student Information System (SIS), with particular emphasis on the Student Course Registration subsystem. By applying enterprise architecture principles commonly used in higher education institutions, this project aims to propose a structured architectural model that supports clearer system alignment, improved information flow, and more effective management of student academic processes at UTHM.

1.2 PROBLEM STATEMENT

This section outlines the key problems that motivate the development of the proposed enterprise architecture for UTHM. It highlights current issues related to the existing enterprise architecture,

associated inefficiencies, and potential risks affecting student-related systems. The discussion focuses on the Student Information System (SIS), particularly the Student Course Registration subsystem, to justify the need for a structured enterprise architecture approach.

1.2.1 Current Issues

UTHM utilizes multiple information systems to support its academic and administrative operations, including student related systems such as SIS. However, the enterprise architecture currently applied is basic and not formally documented as a comprehensive enterprise-wide model. As a result, the relationships between business processes, applications, data, and technology are not clearly defined across the university's core operational systems.

1.2.2 Inefficiencies

Due to the absence of a formalized and standardized enterprise architecture, coordination between systems may be inconsistent. System development and enhancement activities may be carried out independently by different units, leading to challenges in system integration and information flow. These inefficiencies are more apparent in student-related systems, where accurate data handling and process alignment are critical to supporting academic activities such as course registration.

1.2.3 Risks and Limitations

Without a clear architectural structure, the Student Course Registration subsystems under the SIS faces limitations in terms of scalability, and long-term planning. The lack of clear mapping between business requirements and system functions increases the risk of misalignment with UTHM broader academic and digital objectives. In addition, future system improvements may become more complex and harder to manage without a structured enterprise architecture to guide decision making.

1.3 PROJECT OBJECTIVES

The main objective for this project is to design and purpose an enterprise architecture for Universiti Tun Hussein Onn Malaysia (UTHM). The objectives will be focused on analyzing the current system environment, applying a structured enterprise architecture framework which is TOGAF

framework, and supporting future enhancing management of university operations. The objectives of this project are as follows:

1. To analyze the existing information system used by Universiti Tun Hussein Onn Malaysia (UTHM) including academic system, management system and others to gain greater understanding the current system environment and challenges.
2. To design an Enterprise Architecture (EA) for UTHM by using TOGAF framework as a structured guideline to structure the business processes, applications, data, and technology
3. To provide a structured architectural reference for future system improvement to make the system more manageable and systematic

1.4 PROJECT SCOPE

This section defines the project boundaries to ensure that this project is achievable within the given constraints. It clearly outlines the areas that covered by the project (in-scope) and the part that are not included (out-of-scope) particularly related to the proposed enterprise architecture for UTHM.

1.4.1 IN-SCOPE

The in-scope areas of this project include:

- Analysis the existing enterprise systems environment at UTHM such as overview of academic, student, and management-related system.
- Focus on the Student Information System (SIS) specifically the student Course Registration subsystem as main case study
- Design a proposed Enterprise Architecture for UTHM using the TOGAF framework as structured guideline that are covering key architectural layer such as Business, Application, Data and Technology Architecture
- Create an architectural models and documentation as a reference for improvement of the system UTHM in future.

1.4.2 OUT-OF-SCOPE

The out-of-scope areas of this project include:

- Development a system, coding or any related to implementation of proposed enterprise architecture.
- Perform a testing such as security testing and user acceptance testing of existing or proposed system.
- Evaluation of hardware, cost analysis, or budgeting related to system implementation.

1.5 PROJECT IMPORTANCE

The development of a formalized Enterprise Architecture (EA) for UTHM is critical to ensuring the university's digital infrastructure can support its growing academic and operational demands. This project holds significant importance in three key areas:

1.5.1 Strategic Alignment and Resources Optimization

Currently, UTHM's information systems operate in silos, which often leads to redundant spending where different faculties might procure separate software solutions for similar needs. This project is important because it utilizes the TOGAF framework to create a unified blueprint. By centralizing the architecture, the university can identify overlapping resources and streamline IT investments. This ensures that every ringgit spent on technology aligns directly with UTHM's strategic objectives, reducing unnecessary costs, and fostering long-term sustainability.

1.5.2 Operational Efficiency and System Integration

The project addresses the critical issue of fragmentation within UTHM's current environment. By proposing a structured architecture for the Student Information System (SIS), this project eliminates the inefficiencies caused by independent system development. It provides a standard reference for IT teams, ensuring that data flows seamlessly between the academic and administrative departments. This reduces data redundancy, minimizes the risk of integration errors, and simplifies the maintenance of complex university systems.

1.5.3 Enhanced Student Experience

Since the project focuses specifically on the Student Course Registration subsystem, its immediate importance lies in directly improving student experience. Course registration is a high-traffic process that frequently suffers from system congestion. A well-architected system ensures scalability, preventing system crashes during peak registration times. Furthermore, this architectural update aims to streamline the user interface, making the registration process more intuitive and less stressful for students.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter presents a literature review related to enterprise architecture (EA) in universities, providing both theoretical and practical foundations for this study. This chapter begins with EA definition, key concepts, and some applications within higher education institutions. Meutia, Sulistiani, Budiarti, and Sari (2022) state that EA serves as a strategic framework that aligns business processes, information system, and technology infrastructure with organizational objectives, helping universities to manage complex operations and support digital transformation.

Besides, the review also highlights previous studies on EA in universities, emphasizing how EA frameworks have been implemented in universities to coordinate administrative, academic, and technical functions (Hindarto, 2023; Amin, Hussain, Daoud, Abu Bakar, & Ahmad, 2024). In addition, this chapter examines various subsystems, discussing the existing implementation and architectural considerations of these sub-systems. Furthermore, this chapter also explores the technologies used in EA and its sub-systems, including software platforms and architectural frameworks.

Finally, this chapter concludes with a summary of key findings from previous studies to establish a solid foundation for understanding EA in universities worldwide.

2.2 ENTERPRISE ARCHITECTURE

Enterprise Architecture (EA) is a method that an organization uses to plan the business and IT in a way that each will fit together well. Following the TOGAF standard, EA segments the organization into different segments to show how processes, information, applications, and technology support each other (The Open Group, 2018). EA also serves as a clear blueprint to map business goals to their supporting IT systems (Lankhorst, 2017). EA typically includes four sectors which are Business Architecture, Data Architecture, Application Architecture, and Technology

Architecture (The Open Group, 2018). Together, it helps everyone understand how the organization works and how IT supports the organization.

In University, the Data Architecture layer is crucial, as they require high data quality and data organization for all aspects of their business, including students, staff, and research. Previous works indicate that universities often use a TOGAF framework for guidance, but they adjust it to their needs (Pedroso, 2021). Universities focus on management of organizational data, interoperability of organizational systems, accessibility, and security. However, challenges like different departments managing their own systems, legacy and outdated technologies, and a lack of trained EA professionals prevent them from implementing EA effectively. Thus, universities require good planning and teamwork for the successful application of EA.

2.2.1 UNIVERSITY OF LAMPUNG

A study by Nama and Kurniawan (2017) designed an adaptive IT infrastructure based on the enterprise architecture framework The Open Group Architecture Framework (TOGAF) Architecture Development Method (ADM) for the University of Lampung (Unila). The Academic Information System (Siakad) used by Unila, which has been operational for a long time, is not integrated with other applications such as finance, employee, and planning systems. This causes delays in gathering data when leaders or stakeholders request reports across the work unit.

Figure 2.1 shows the enterprise architect that was proposed to Unila. According to the study, the IT unit should be in charge of centralizing infrastructure management. In order to deliver a scalable, dependable, and flexible service, the unit must also implement security rules, data management and distributed system services and run the Disaster Recovery Center.

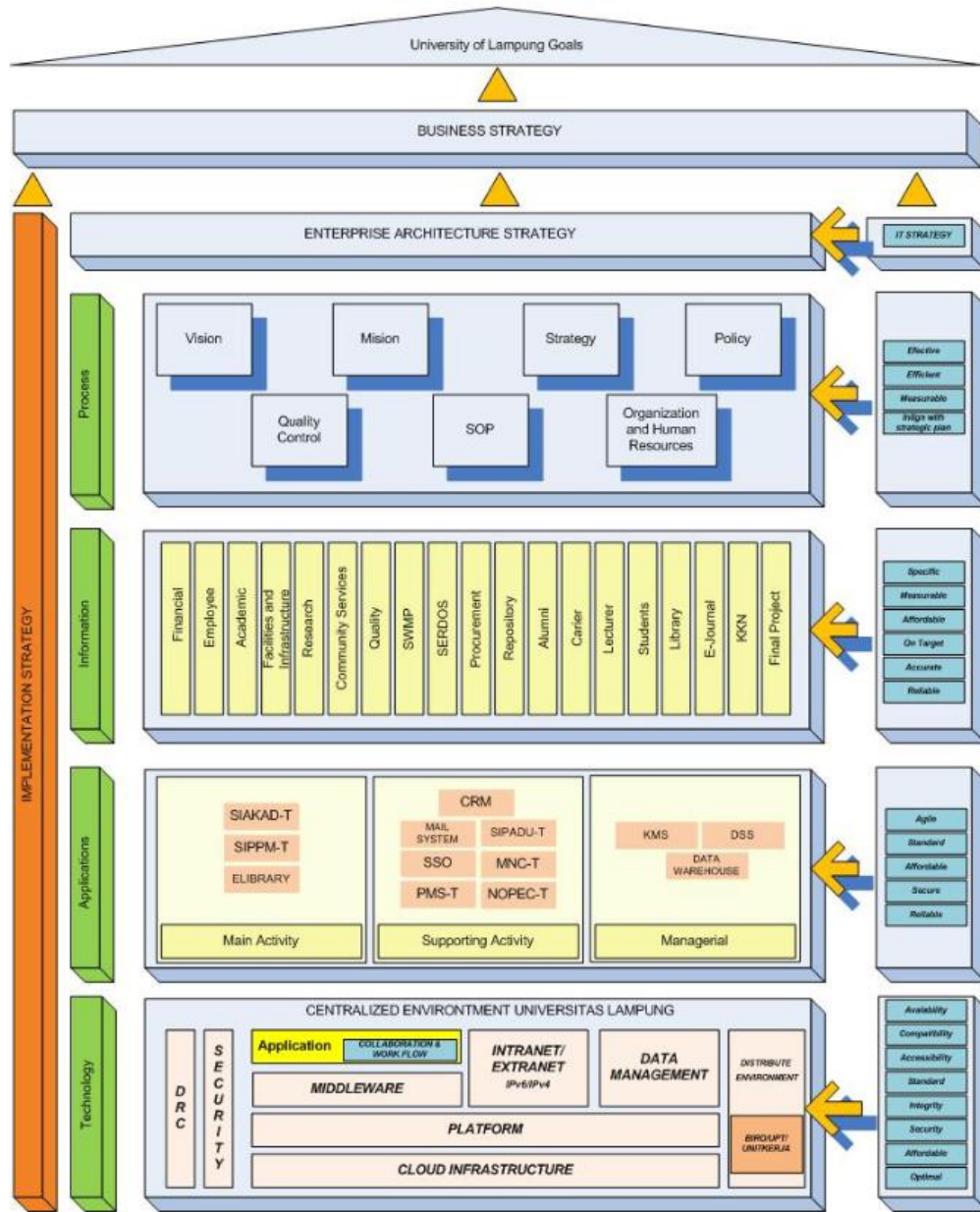


Figure 2.1 Enterprise Architecture proposed to Unila. (Nama & Kurniawan, 2023)

2.2.2 NATIONAL UNIVERSITY SOUTH JAKARTA

Another study by Hidarto (2023) investigates the importance of EA in enabling digital transformation of the management systems in universities. The paper uses The Open Group Architecture Framework (TOGAF) framework as its research method, as it provides a methodical framework for the development, organization, execution, and oversight of enterprise information technology architecture. The paper defined digital transformation as the migration from manual processes to using application systems that can automate those processes, as universities often face problems such as human errors and slow information flow and data retrieval from manual processes. The paper also highlights opportunities using application systems such as access to advanced technologies, artificial intelligence (AI), big data analytics, and the Internet of Things (IoT) that can increase effectiveness in operation, boost the process of teaching and use resources efficiently.

Figure 2.2 shows the enterprise architecture described by the paper. The Integration Services establishes a connection to multiple sources, allowing data integration from external systems using API. The portal is a central interface through which teachers, staff, and students can combine their access to information and university resources. Data & Analytics oversees collecting, analysing, and using data to better decision-making, while the Back-Office systems manage the internal operations of the university.

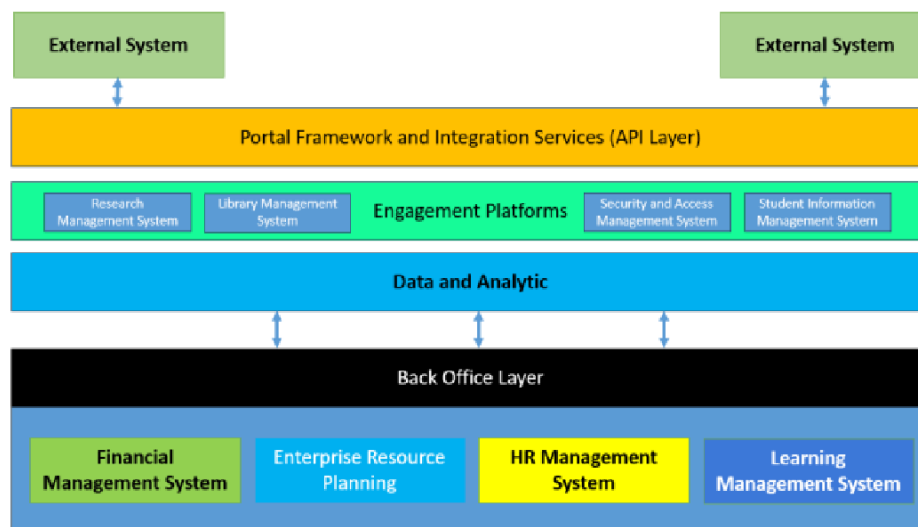


Figure 2.2 Enterprise Architecture by Hidarto. Hidarto (2023)

2.2.3 NEW YORK UNIVERSITY

Figure 2.3 shows the New York University's (NYU) Enterprise Architecture operating model where it explains how the university connects its overall strategy with IT planning and governance. NYU uses EA to connect key areas such as learning and teaching, research, community engagement, and administration with its IT and digital systems. This shows that EA is not only about technology, but it also about supporting the university's main goals. By aligning EA with NYU's strategic, the university ensures that IT investments and digital initiatives support teaching, research, and daily operations across different campus.

At the architecture level, NYU's EA covers business process, data, application and technology infrastructure. Important data such as student records, research information, and financial data are managed clearly to support reporting, better decision making, etc. Core system like the Student Information System (SIS), Learning Management System (LMS), and identity services are shared across the university to avoid any duplication in the system. In addition, technology standards such as cloud, security governance, and infrastructure guidelines help support both academic and research activity. Overall, New York University Enterprise Architecture shows how strong governance can help universities manage a complex system while keeping IT decisions match with institutional goals, making it useful reference for other universities.

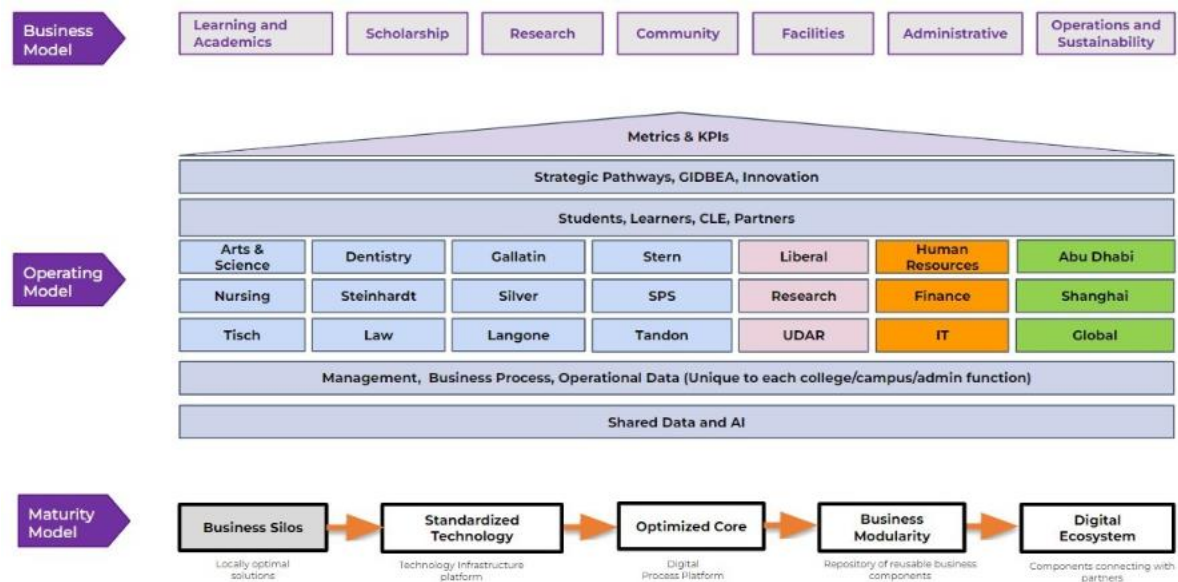


Figure 2.3: Enterprise Architecture New York University

2.3 RELATED SUB-SYSTEM

Research has been done to identify the subsystems used in university by analyzing them through the types of Enterprise Architecture methods and technologies. These studies look at the implementation and integration of systems like the Human Resources Management System (HRMS), Academic Information System (AIS), Student Information & Subject Registration System (SIS) and Learning Management System (LMS) and Business Intelligence (BI) and Data within EA frameworks to support coordinated processes, data consistency and higher education institutions' operational and strategic goals.

2.3.1 Human Resources Management System (HRMS)

An HRMS is a system that combines information technology and human resources management (HRM) to automate and support HR procedures. For instance, it manages payroll, training, performance reviews, resignations, resume (applicant), employee details and attendance. The goal is to increase operational efficiency in HR departments, decrease manual work and make data access more dependable and simpler (Navaz et. al, 2013).

According to Sikarwa, 2018, the primary goal of HRMS is to guarantee that the organization's strategic goals can be met and the universities' needs can be satisfied in terms of both quantity and quality. The method used from the study is the Open Group Architecture Framework (TOGAF) 9.1 where it stated at preliminary phase; in addition to managing the standard data sources in universities like payroll, employee attendance, hiring and leave, HRMS also handles the demands of promotion management, period payment increase analysis, reward and punishment management and staff education. In particular, HRMS in universities mainly covers these functional areas including payroll management, human resources management, institutional management, civil service management, education and training management, performance management and function management. Furthermore, universities have a HRMS integrated system design that can be integrated with current systems so that its HRMS architecture may supply information required by end users as well as the business needs of universities in managing human resources.

2.3.2 Academic Information System (AIS)

A university's Academic Information System (AIS) is a software program that keeps track of all academic data including timetables, grades, courses and student information. Students can access their transcripts, study result cards and academic information online while administrators can effectively manage student records, enter grades, manage courses and create reports. It replaces manual, paper-based procedures, saves time, minimizes mistakes and improves staff and student access to academic management (Manuhutu et. al, 2018).

According to Astri and Gaol, 2013, the academic information system manages the school's academic processes from new student entrance to graduation ceremonies. In the study, this system performs essential functions such as managing admission test results, scheduling, curriculum, grade management, student administration and academic report applications. The study uses the Enterprise Architecture Planning (EAP) methodology to connect technology, data and apps across the systems. Hence, better administration, coordination and long-term system planning are made possible by the school's ability to create a comprehensive blueprint through EAP that synchronizes its academic information systems with an integrated enterprise-level architecture.

2.3.3 Student Information & Subject Registration System (SIS) and Learning Management System (LMS)

The Student Information System (SIS) and the Learning Management System (LMS) constitute the core technological foundation that supports both academic administration and instructional delivery within a university. In contemporary digital transformation initiatives these systems are expected to function not as isolated components but as fully integrated parts of a comprehensive University Management System (UMS) (Hindarto, 2023).

The SIS is responsible for handling the complete lifecycle of student information. This includes initial registration, enrollment in subjects and the maintenance of academic records, and the management of graduation processes. Earlier forms of SIS frequently encountered issues such as fragmented datasets and duplicated entries which resulted in inconsistencies across institutional records (Lamey et al., 2023). To address these challenges many institutions, rely on EA frameworks that are often modelled on methodologies such as TOGAF. These frameworks enable

the design of more coherent processes, unified databases, and stronger linkages to subsystems that include finance and scheduling (Metia et al., 2022).

The LMS complements the SIS by managing instructional resources and assessments. Effective academic operations require that the LMS maintain seamless integration with the SIS to ensure the accurate synchronization of enrollment data and student performance records. Lamey et al.(2023) emphasize that the development of intelligent and integrated academic solutions depends on consistent and harmonized data shared across both systems. Case studies demonstrate that EA planning (Nama and Kurniawan, 2017) plays a central role in shaping the Application Architecture layers to support reliable academic data flows.

2.3.4 Business Intelligence (BI) and Data

The Business Intelligence (BI) and Data Subsystems represent the strategic dimension of Enterprise Architecture implementation. Its purpose is to transform institutional data into reliable insights that guide decision making, policy information, and overall governance. A central objective of EA within digital transformation initiatives is to advance universities toward a fully data driven operational environment (Hindarto, 2023).

The effectiveness of any BI subsystem is determined by the strength of the Data Architecture established during the EA planning process. This stage involves the standardization of data definitions and the assurance of data quality throughout all primary source systems such as the SIS, HRMS, and FMS before these data are transferred into a centralized data warehouse (Meutia et al., 2022). Without rigorous data governance the outputs produced by BI platform risk becoming fragmented or unreliable.

According to Lamey et al. (2023) advanced analytical and intelligent institutional systems require a well-integrated and dependable data foundation. Such a foundation allows universities to perform tasks that include forecasting student retention, allocating resources efficiently, and monitoring institutional performance indicators. Consequently, the application of EA principles (Nama and Kurniawan, 2017) extends beyond merely connecting technological subsystems. It involves structuring the data environment in a way that supports meaningful strategic intelligence.

2.4 TECHNOLOGY USED

This section serves to describe and justify the key technological platforms, frameworks, and tools selected for the implementation of the enterprise system under study. This part highlights the infrastructure and digital solutions chosen for their relevance, effectiveness, or innovation, providing a clear rationale for why these technologies suit the project's goals and context. In this case of university enterprise systems, this section explains how foundational technologies like SAP Business Technology Platform (BTP) support the digital transformation of campus services and operations, allowing for centralized management, real time analytics, and seamless integration across diverse academic and administrative functions.

2.4.1 King Abdullah University of Science and Technology (KAUST)

The project uses the SAP Business Technology Platform as a central technological enabler, similar to the real-life implementation at King Abdullah University of Science and Technology. Building on SAP BTP, KAUST created an integrated digital platform called "KAUST Central," which consolidated the essential campus services-dining, transportation, security, payments, and communication-into one mobile and web application. The system is exemplary, depicting how SAP BTP can enable a university-wide digital transformation through scalable cloud services, intelligent integration, and user-centered design.

Several SAP BTP components were leveraged for the KAUST project. SAP Mobile Services SDK was used to build native apps for iOS and Android, thus allowing real-time access to campus functions. SAP Analytics Cloud allowed data-driven decisions based on tracking student behavior and providing personalized content, while SAP Experience Management (Qualtrics) continuously gathered user feedback for refining campus services. Moreover, the SAP BTP cloud infrastructure enabled seamless integrations with legacy systems and third-party applications through standardized APIs, which is very important in a distributed university setting where various departments operate on different platforms.

From an enterprise architecture perspective, with SAP BTP, KAUST has implemented a modular and flexible architecture scaling across departments while preserving centralized governance. Single sign-on simplified access to the system, while the cloud-native setup provided support for

rapid updates and extensions, such as adding COVID-19 information features and smart city modules. Most impressively, the platform has managed to cut campus services' wait times from 10 minutes down to 40 seconds and allowed logistics, dining, and transport on campus to be coordinated through real-time dashboards and automated processes.

The use of SAP BTP at KAUST represents one way in which universities are moving beyond traditional ERP models toward a campus-as-a-platform model. It confirms SAP BTP as appropriate, future-ready technology for university enterprise systems, which supports digital innovation, integration, analytics, and smart campus ecosystems.

2.5 SUMMARY

The literature review highlights the importance of Enterprise Architecture (EA) in helping universities to align their business procedures, data management, applications, and technology infrastructure to meet institutional goals and digital transformation. Previous studies show that universities often adopt EA frameworks such as TOGAF framework, but universities usually adjust the framework so that they could fit their own needs, especially when dealing with issues such as decentralized departments, out-dated systems, and diverse stakeholder needs.

Case examples from University of Lampung demonstrate how EA can support better alignment between strategy and IT, improve data management, integrate applications, and set clear technology standards. Meanwhile, insights from the National University South Jakarta emphasize that universities often change from manual processes to digital transformation using application systems that can automate those processes.

In addition, the review of key university sub-systems, including HRMS, AIS, SIS, LMS, BI, and Data shows how EA helps universities ensure smooth data flow, reduce duplication, and design scalable systems. Furthermore, the use of technologies, such as SAP Business Technology Platform (BTP) as demonstrated by KAUST, further show how cloud-based platforms enable universities to deliver smart campus services and support long-term digital innovation.

In conclusion, the literature establishes EA as a strong foundation for modern university enterprise systems, providing both strategic direction and technology integration to enhance overall institutional performance.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

This project aims to design an enterprise architecture for Universiti Tun Hussein Onn Malaysia (UTHM). To achieve this goal, the Waterfall development approach is adopted in this project. The methodology guides the development process of this project from starting to the end, which is planning, analysis, design, implementation, and testing the enterprise architecture components. By following the methodology, the project ensures that every phase of the methodology was implemented successfully and aligned with UTHM operational requirements.

3.2 CHOSEN METHODOLOGY

The Waterfall methodology is a sequential development process where every phase must be completed before moving on to the next phase. It is suitable to be applied in projects that have clear and fixed requirements.

The reason why Waterfall methodology is chosen is because of its structured flow. Since enterprise architecture involves complex systems and long-term planning, understanding the correct requirements is very important. With this methodology, all requirements must be clearly understood before moving to next phase to design the real enterprise architecture.

Other than that, each phase in the Waterfall approach such as requirements analysis, design, implementation, and testing is clearly documented. These documentations will help the university a lot in future maintenance or enhancement.

Moreover, since the phases are clearly defined, it is easier to manage resources, manpower, and project schedules. This reduces uncertainty of the project and reduces the risk of project failure. Therefore, the Waterfall methodology is a suitable approach for this project.

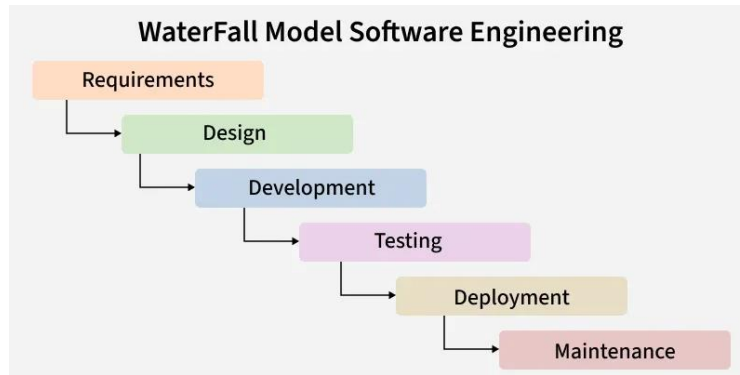


Figure 3.1 Waterfall Methodology Phases (GeeksforGeeks, 2025)

3.3 PHASES OF METHODOLOGY

As illustrated in Figure 3.1, this project is led by the six distinct phases of the Waterfall Model. Each phase depends on the deliverables of the previous one, making sure of a logical progression from recognizing the fragmentation issues at UTHM to delivering an interrelated Enterprise Architecture blueprint.

3.3.1 Requirements

The first phase involves gathering and analysing the necessary information to understand the current state of UTHM’s business and IT environment specifically identifying pain points caused by the separation of various departmental systems. In this stage, the project team conducts data gathering to understand the existing siloed systems across different faculties and administrative units, defining the scope for unifying these departments into a single platform. The outcome of this phrase is a comprehensive Problem Statement and User Requirement Specification (URS) that clearly identifies the gaps in the current architecture.

3.3.2 Design

Based on the requirements gathered, the team moves to the design phase to create the architectural blueprints that define the “To-Be” state of the organization. The team adopts the TOGAF (The Open Group Architecture Framework) approach to model the target architecture including the Business, Data, Application, and Technology layers required to support a unified university platform. This phase concludes with the production of high-level architectural diagrams and a proposed Enterprise Architecture design document that solves the issues identified in the analysis

phase. technologies were defined in this phase in order to design the system architecture, which was done by using SAP Build Apps. Each member was assigned different modules and was assigned to make visual representations of the modules using UML diagrams such as use case diagrams, ERD diagrams, use case descriptions, activity diagrams, and sequence diagrams. Interface design was also provided to give an overview of how the finished product would look.

3.3.3 Development

During the development phase, the system was constructed with SAP Build Apps, and the process began with a structured implementation of the data layer, building the necessary entities based on each member module. Once the data foundation was established, UI design and logic implementation were completed, with functional modules customized to each module.

3.3.4 Testing

This phase focused on testing the quality of the custom logic and data bindings, beginning with unit testing to ensure that the complex lookup formulae for dynamic session filtering produced exact data from the database. Functional testing was then performed using the SAP Build Apps Previewer, where each use case module was tested to ensure that the system appropriately handled both success and failure scenarios, such as mismatched passwords or wrong current credentials. Finally, integration testing verified that the data layer and user interface stayed in sync, guaranteeing that changes to one page of the system's information were promptly reflected in the application's local memory. The predicted and actual results were then compared to ensure that all functional requirements were properly met.

3.3.5 Deployment

Once the design is verified, the architecture is ready for delivery to the stakeholders. The final Enterprise Architecture proposal is formally presented to the university management, marking the transition from planning to potential adoption. For this project, this phase represents the submission of the final project report and the formal presentation of the unified platform proposal.

3.3.6 Maintenance

The final phase focuses on the long-term sustainability of the architecture. Although this project concludes upon submission, this phase outlines recommendations for future governance to ensure the unified platform remains adaptable to new educational technologies or changes in UTHM's strategic direction. The key output here is a set of recommendations for future architecture governance and scalability.

3.4 PROJECT PLANNING SCHEDULE

The project planning schedule, as illustrated in Figure xx, follows the sequential stages of the Waterfall methodology to ensure a structure development process. The timeline begins with a collective planning and analysis phase, where the entire team conducts a site visit to UTHM and prepares interview questions to define the system requirements. The project moves into design phase, where individual responsibilities are assigned to all team members to develop specific tasks for the seven modules. The schedule then transitions into the implementation phase, focusing on database creation, and coding. Finally, the process concludes with a testing and deployment phase.

TASKS	ASSIGNED TO	DURATION (WEEKS)	27/10-31/10	3/11-7/11	10/11-14/11	17/11-21/11	1/12-5/12	8/12-12/12	15/12-19/12	22/12-26/12	29/12-2/1	5/1-9/1	12/1-16/1
Meeting with group members	ALL	1											
Project planning	ALL	1											
Requirements													
Overview and Problem Statement	SYAHILI	2											
Project Objectives and Project Scope	NAIM	2											
Project Important	ADRIANA	2											
Literature Review, EA concepts, explore SAP BTP	ALL	2											
Analysis													
Prepare Interview Questions	ALL	1											
Site Visit to UTHM	ALL	1											
Define System Requirements	FARRA	1											
Company Organization Structure	SAFIYA	1											
Analysis Current System	CHUA	1											
Design													
EA AHS & TO-BE	ALL	2											
System Architecture	IMAN	2											
Use Case Diagram and Description													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4 : Student Course Registration Module	CHUA	1											
Process 5 : Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHILI	1											
Process 7 : Course Approval Module	SAFIYA	1											
Activity diagram													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4 : Student Course Registration Module	CHUA	1											
Process 5 : Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHILI	1											
Process 7 : Course Approval Module	SAFIYA	1											
Sequence diagram													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4 : Student Course Registration Module	CHUA	1											
Process 5 : Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHILI	1											
Process 7 : Course Approval Module	SAFIYA	1											
Database Design													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4 : Student Course Registration Module	CHUA	1											
Process 5 : Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHILI	1											
Process 7 : Course Approval Module	SAFIYA	1											
Interface Design													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4 : Student Course Registration Module	CHUA	1											
Process 5 : Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHILI	1											
Process 7 : Course Approval Module	SAFIYA	1											
Development													
Create Database													
Process 1 : View Student List for each Section Module	FARRA	2											
Process 2 : Credit Hour Module	NAIM	2											
Process 3 : Section Scheduling Module	ADRIANA	2											
Process 4 : Student Course Registration Module	CHUA	2											
Process 5 : Manage Student Profile Module	IMAN	2											
Process 6 : Section Allocation Module	SYAHILI	2											
Process 7 : Course Approval Module	SAFIYA	2											
Coding of system's main function													
Process 1 : View Student List for each Section Module	FARRA	2											
Process 2 : Credit Hour Module	NAIM	2											
Process 3 : Section Scheduling Module	ADRIANA	2											
Process 4 : Student Course Registration Module	CHUA	2											
Process 5 : Manage Student Profile Module	IMAN	2											
Process 6 : Section Allocation Module	SYAHILI	2											
Process 7 : Course Approval Module	SAFIYA	2											
Testing & Deployment													
EA Presentation	ALL	1											
Final Project Demo	ALL	1											

Figure 3.2: [Gantt Chart](#)

3.5 SUMMARY The enterprise architecture for Universiti Tun Hussein Onn Malaysia (UTHM) is developed using the Waterfall methodology, which follows a structured and sequential development process. This approach organizes the project into clear phases, including planning, analysis, design, implementation, and testing, ensuring that each stage is completed before moving forward and remains aligned with UTHM operational requirements.

The Waterfall methodology is suitable for this project because enterprise architecture involves complex systems, and long-term planning that require clearly defined requirements. Understanding all requirements at the early stage helps reduce the risk of system misalignment and improves the quality of architectural design. In addition, clear documentation produced in each phase supports future system maintenance and enhancement activities.

The project planning schedule follows the sequential flow of the Waterfall model, beginning with site visits and requirement analysis. The process then moves into the design phase, where responsibilities are distributed among team members, followed by implementation activities such as database creation and the use of framework such SAP Build. The project concludes with testing and deployment to ensure the proposed enterprise architecture functions as intended within the university environment.

CHAPTER 4

ANALYSIS AND DESIGN

4.1 INTRODUCTION

This chapter's focuses on the design and analysis of the proposed system and the Enterprise Architecture (EA) for Universiti Tun Hussein Onn Malaysia (UTHM). It looks at the current system to find issues and requirements followed by the right enterprise architecture and system design approaches to create an improved EA. This chapter's outputs include the system architecture, database design and interface design to offer a clear foundation for the EA implementation stage.

4.2 COMPANY ORGANIZATION STRUCTURE

The Information Technology Centre (PTM) at Universiti Tun Hussein Onn Malaysia (UTHM) operates under a structured hierarchy led by a Director and a Senior Deputy Director, supported by an Office Secretary Assistant. The department is organized into six main divisions that handle different parts of the university's technology needs. The Application Management Division is responsible for the Application Development Unit and the Student and Staff Information System Unit. Infrastructure and Cyber Security handles data centre, networks, and telecommunications.

The Digitalization Technology, which handles web and multimedia services, and ICT Facilities and Support, which manages technical help and campus equipment. To ensure projects follow university rules, the Governance, Strategic, and ICT Training oversees planning and staff training. Finally, the Administration and Finance division manages the department's budget and office operations.

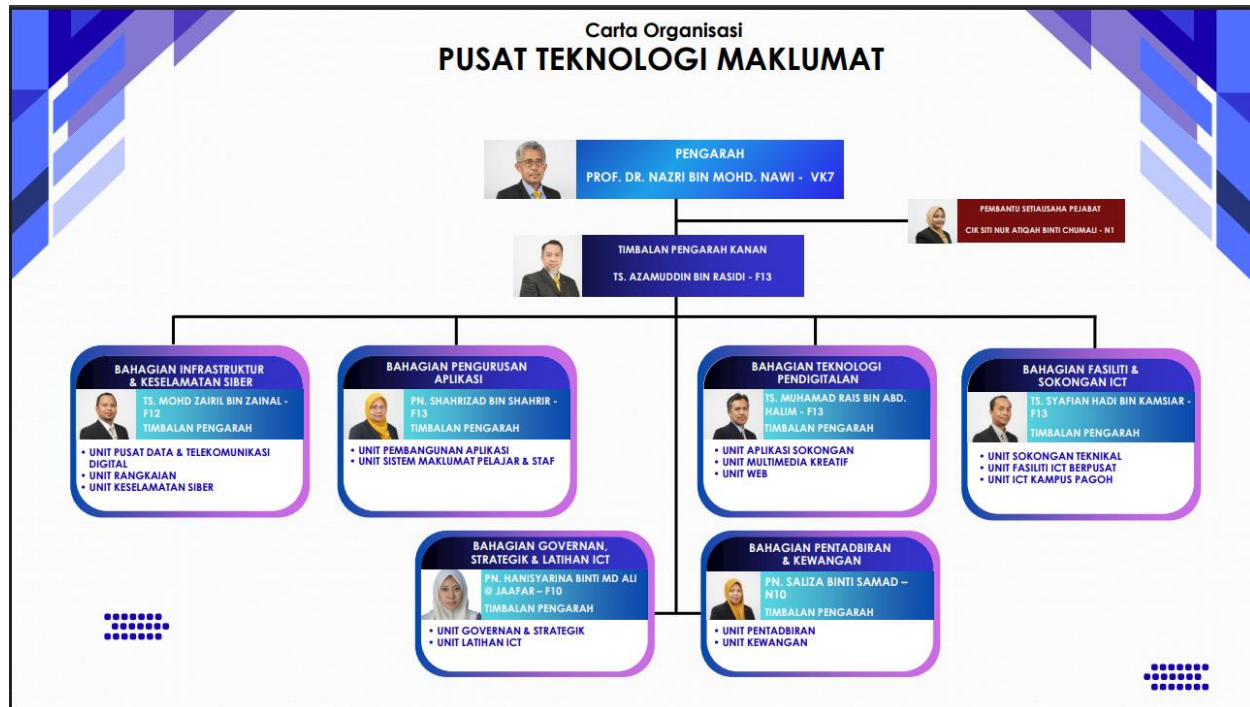


Figure 4.1 Organization Chart of Information Technology Centre (PTM) at Universiti Tun Hussein Onn Malaysia (UTHM)

4.3 CURRENT SYSTEM ANALYSIS

UTHM currently uses a combination of limited manual processes and computerized systems. Their digital operations are supported by a Tier 3 infrastructure. The data center uses Data Structure Infrastructure Management (DCIM) monitoring, air-based condenser cooling system and hot and cold aisle containment to track performance and reduce power consumption. Additionally, they have a staging area where vendors set up systems before entering the containment area.

Furthermore, UTHM's primary digital technologies include a central student database and Learning Management System (LMS) for teaching materials. These systems are not fully integrated so the majority of data transfers are carried out manually one at a time especially when the format or structure varies between the systems. However, API connections are used to manage big data transfers to increase efficiency and lower errors. The process was also carried out for data replication to the Data Recovery Center (DRC) where data is transmitted for disaster recovery to

a DRC in Pagoh. Moreover, Single Sign-On (SSO) was implemented to make system access easier but frequent user concerns like forgotten password and Wi-Fi login issues still happen.

Additionally, UTHM is transitioning from old systems to a new web-based integrated system called Total Campus Integrated System (TCIS). This migration is ongoing and being completed phase by phase based on modules. Even though UTHM wants to be entirely digital by 2030, some administrative tasks are still done manually because of outdated equipment and legacy systems. They stated that some hardware comes with outdated operating systems which makes it less compatible with modern technologies. In addition, UTHM uses a WAF (Web Application Firewall) to secure web portals and websites as well as two firewalls which filter threats often from China and another from Europe to protect the system and data.

Overall, UTHM is actively pursuing digital transformation and has built a strong infrastructure. However, system integration, manual data handling, legacy equipment and user access continue to be challenges in achieving full end-to-end digital transformation by 2030.

4.4 SYSTEM REQUIREMENTS GATHERING

System requirements gathering is the process by which we gather data to understand user needs and the enterprise system that will be developed. It helps us to determine what functions, features and improvements are needed. Two methods were used to collect data for this industrial visit which are questionnaire and interview. Before visiting UTHM, a set of Enterprise Architecture related questions was prepared based on the TOGAF elements. Moreover, we asked the presenter directly during the talk session in order to gather more information. Below are the sample questions that were presented using the TOGAF elements:

1. Business Architecture

- What are the university's primary business capabilities (e.g., teaching, research, finance, HR)?
- What is the top 10 business processes that require transformation?
- Which processes are manual, duplicated, or inefficient in the current environment?

- What future state vision or strategic goals does UTHM want the new EA to support?
- What are the critical KPIs (student success, research productivity, operational efficiency)?

2. Data Architecture

- What core data entities exist (students, staff, courses, finance, research, facilities)?
- What data standards or naming conventions does UTHM currently use?
- Where is data duplicated across systems?
- Is there single master database in UTHM for student data or separate with student database and learning database?
- What current workflow transferring between student data to LMS from the master database?

3. Application Architecture

- What are the current enterprise systems in use (student system, finance system, LMS)?
- What functions do these systems provide today?
- What functions are missing?
- What is the most common complain receive from student for current online portal?
- What is the desired future application landscape (modular, unified, cloud-based)?

4. Technology Architecture

- What is the current hosting model (on-premises, cloud, hybrid)?
- Which operating systems, databases, and middleware solutions are used?
- What hardware or infrastructure is aging or needs replacement?
- What is the preferred technology stack for the new system?
- What precaution steps will be taken if electric is cut off?

5. Integration / Inoperability Architecture

- What systems need to integrate with each other (student system ↔ LMS ↔ finance)?
- What integrations currently exist, and what issues do they have?
- What is the strategy for data migration?

6. Security Architecture

- What are the authentication requirements (SSO, MFA, LDAP/Active Directory)?

7. Information / Content Architecture

- How are files stored today (file servers, Google Drive, departmental systems)?

8. Infrastructure Architecture

- How many data centres does UTHM operate?
- What backup and recovery processes are in place?

9. Governance Architecture

- Who makes decisions about IT investments and architectures?
- Does the university need EA governance tools (e.g., architecture repository)?
- Who will be responsible for reviewing or approving the system?

10. Change Management & Implementation Architecture

- How ready are stakeholders for digital transformation?
- What departments are most resistant to change?

4.5 SYSTEM DESIGN

This section describes the design of the proposed system based on the requirements identified during the analysis phase. The system design is presented using enterprise architecture models, system architecture, UML diagrams, database design and interface design to clearly illustrate how the system is structured, user-friendly and capable of improving the efficiency of the existing processes at UTHM.

4.5.1 ENTERPRISE ARCHITECTURE EA – AS IS

Figure 4.2 illustrates the AS-IS enterprise architecture of UTHM. The figure was designed based on the requirements gathered during an interview session with their representatives. The architecture was designed according to their business goals, limitations and constraints specified by them during the interview. Each component of the EA will be discussed on the following sub chapters.

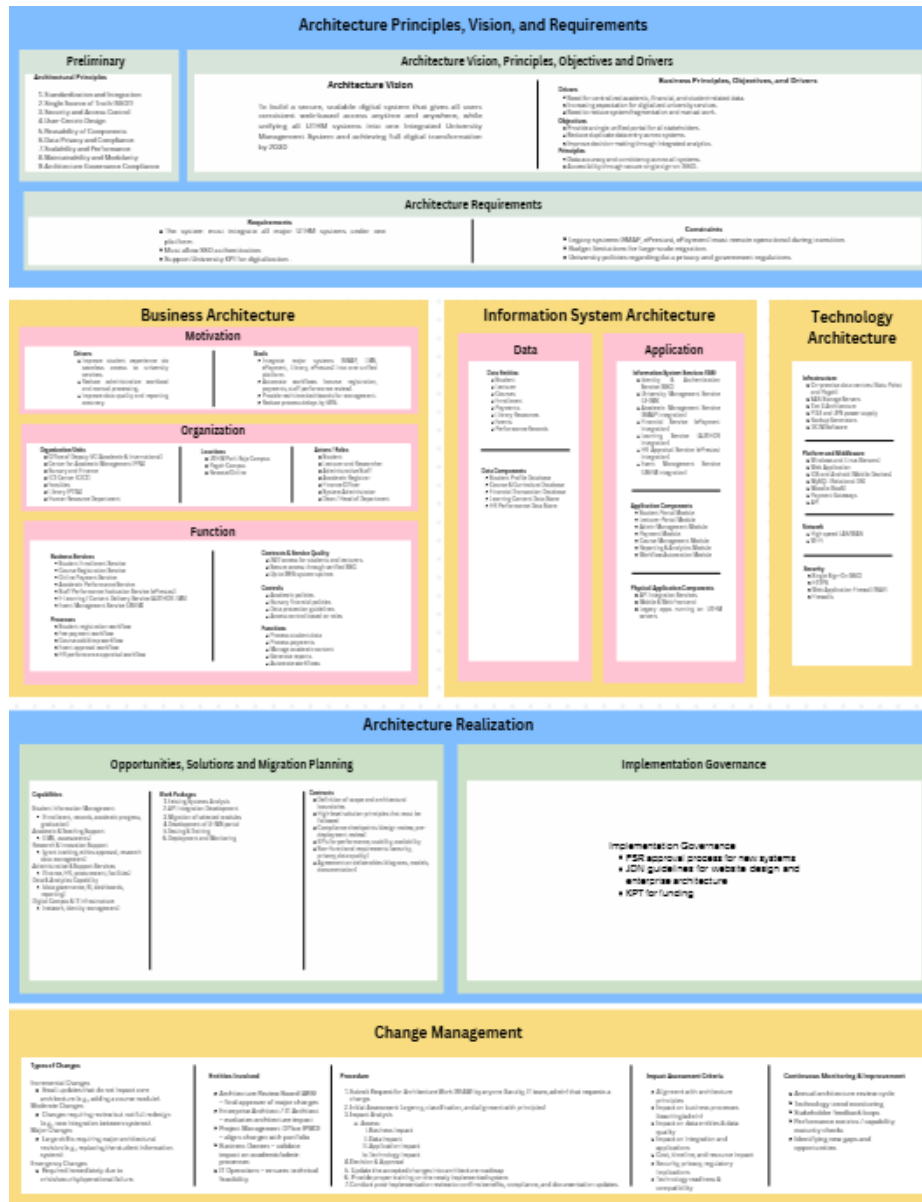


Figure 4.2 Enterprise Architecture AS-IS

4.5.2 COMPONENT EXPLANATION OF ENTERPRISE ARCHITECTURE EA – AS IS

Figure 4.3 shows the diagram that illustrating the architectural principles, visions and requirements. The primary goal of this design is to accomplish complete digital transformation by 2030. This is built on a goal to create a secure, scalable, and unified Integrated University Management System that offers all users uniform web-based access from anywhere. Several business reasons are driving this change, the most important of which is the necessity to combine distributed academic and financial data, as well as the growing demand for digital services. UTHM hopes to provide a single interface for all stakeholders, minimizing manual effort and boosting decision-making through integrated analytics.

To ensure that the system has a strong structure, the preliminary phase develops nine basic architectural concepts. These serve as the "base rules" for development, stressing standardization and integration, as well as a single source of truth (SSOT) to avoid data silos. The principles also focus the user experience through user-centric design while ensuring long-term sustainability through maintainability, modularity, and reusability. Security is a non-negotiable component, with a particular emphasis on data privacy, compliance, and access control, to ensure the architecture stays controlled and legally sound.

The current Architecture Vision (AS-IS) is primarily driven by operational stability and department independence rather than a unified enterprise strategy. It focuses on maintaining the functionality of distinct legacy systems like SMAP and ePayment to ensure daily academic and administrative continuity, relying on established on-premises infrastructure and manual data synchronization to meet immediate, localized business needs without disrupting ongoing university operations.

The effective execution of this vision is dependent on achieving architecture requirements, such as integrating all key existing UTHM systems and implementing Single Sign-On (SSO) for smooth user authentication. However, these objectives must be balanced against real-world constraints. Developers must guarantee that essential legacy systems, such as SMAP, ePrestasi, and ePayment, remain fully functioning during the transition phase. Furthermore, the change must be completed within the constraints of the university's budget and tight government data protection requirements.

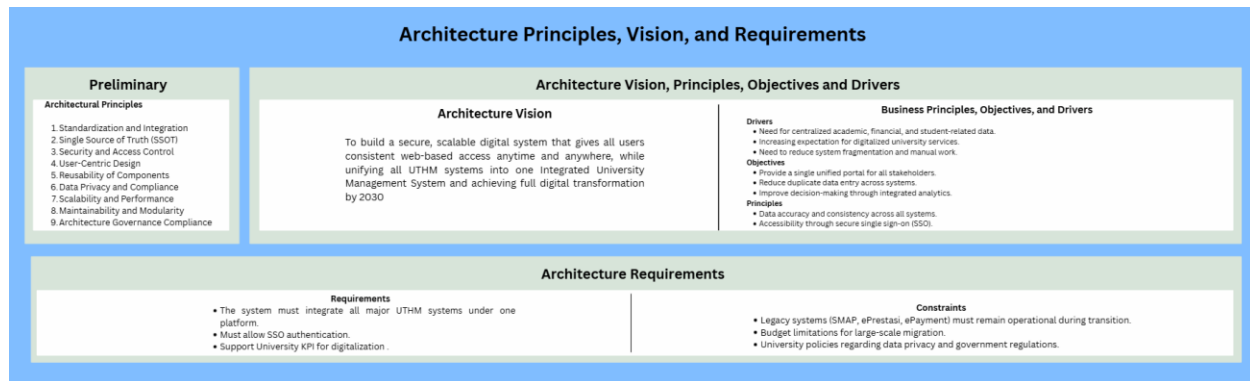


Figure 4.3 Preliminary and Architecture Vision element

Figure 4.4 illustrates the business architecture, information systems architecture and the technological architecture in the EA. The business architecture layer is the foundation, outlining the motivation, organization, and function of the university's digital transformation. It is driven by its goal to improve student experiences while reducing administrative costs by integrating significant systems such as SMAP and LMS. An important performance aim is to cut process delays by 50% while maintaining 99% system uptime. This layer distinguishes between different organizational units, such as the ICT Center and the Bursary, and describes the responsibilities of individuals ranging from students and lecturers to system administrators. Functionally, it defines basic services such as student enrollment and fee collection, which are guided by academic and financial rules to ensure high standards of service.

This information system architecture defines the gap between business needs and technological execution by focusing on data and application components. Data architecture specifies the important entities such as students, lecturers and courses, in addition to the underlying data components, such as the student profile and financial transaction databases, which contain this information. The application architecture describes the Information System Services (ISS), which include the University Management Service (U-IMS) and particular modules for students and teachers. It focuses on the integration of historical systems (such as ePayment and ePrestasi) via an API integration layer, guaranteeing that disparate software components may efficiently communicate under a single site.

The Technology Architecture defines the physical and logical infrastructure needed to support the above levels. The infrastructure is created on-premise data centers in Batu Pahat and Pagoh, with Tier 3 architecture and SAN storage servers to provide high availability. The Platform and Middleware section includes information about Windows and Linux platforms, MySQL databases, and iOS and Android mobile compatibility. This complete stack is linked by high-speed LAN/WAN and Wi-Fi networks. The design protects university data with powerful security methods such as Web Application Firewalls (WAF), HTTPS encryption, and Single Sign-On (SSO).

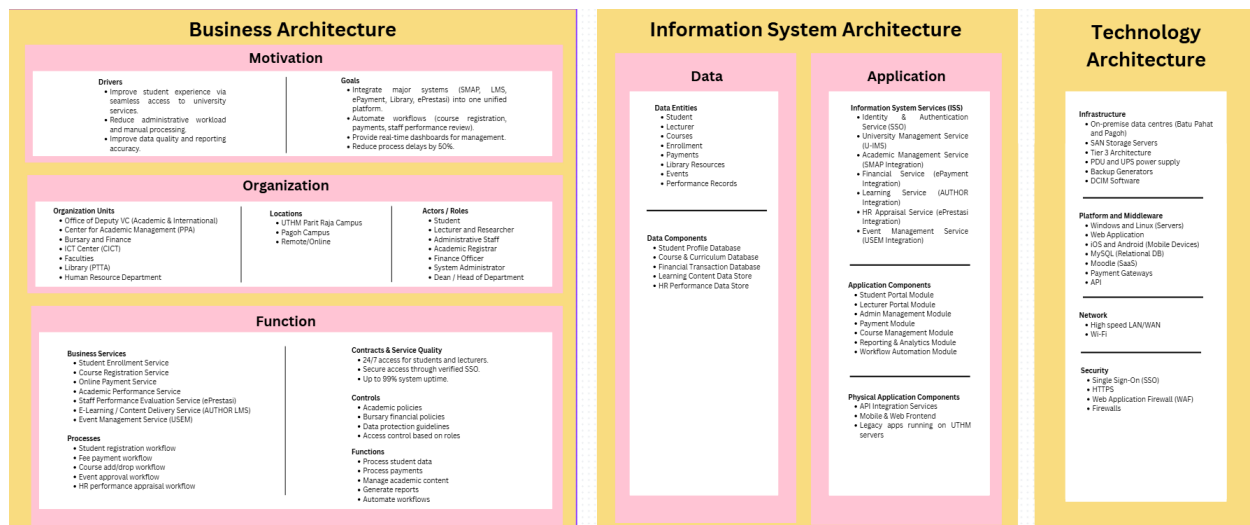


Figure 4.4 Business, Information System and Technology Architecture element

The below EA AS-IS diagram shows the current situation of UTHM's enterprise systems. At present, UTHM already has important capabilities such as Student Information Management, Academic Support, Administrative Services, data & Analytics, and IT Infrastructure. However, many of these systems still work separately. Some processes depend on manual work, and system integration is limited. Because of this, data sharing between systems can be slow, and it is harder for the university to monitor performance and make quick decision. This situation creates an opportunity to improve efficiency and reduce system duplication.

From the migration planning perspective, the AI-IS stage focuses on basic analysis and preparation. The work packages mainly involve existing system analysis, initial API integration, and testing. These steps help UTHM understand which systems need improvement and how they can be connected. The contracts section highlights the importance of defining system boundaries, security requirements, and performance indicators before any major changes are made. Overall, the AS-IS architecture helps UTHM identify weaknesses in the current environment and prepares the university for a more structured transformation.

As shown in Figure 4.5, the Implementation Governance framework for UTHM's current enterprise architecture is defined by a structured oversight mechanism designed to ensure that all technical developments align with both internal policies and national standards. The governance process is anchored by three primary pillars which are the FSR approval process for the validation of new systems, the JDN (Jabatan Digital Negara) guidelines which standardize website design and enterprise architecture practices, and KPT (Ministry of Higher Education) for the strategic oversight of project funding. This framework ensures that any architectural changes, whether incremental or major, remain compliant with institutional mandates while maintaining the operational integrity of critical legacy systems.

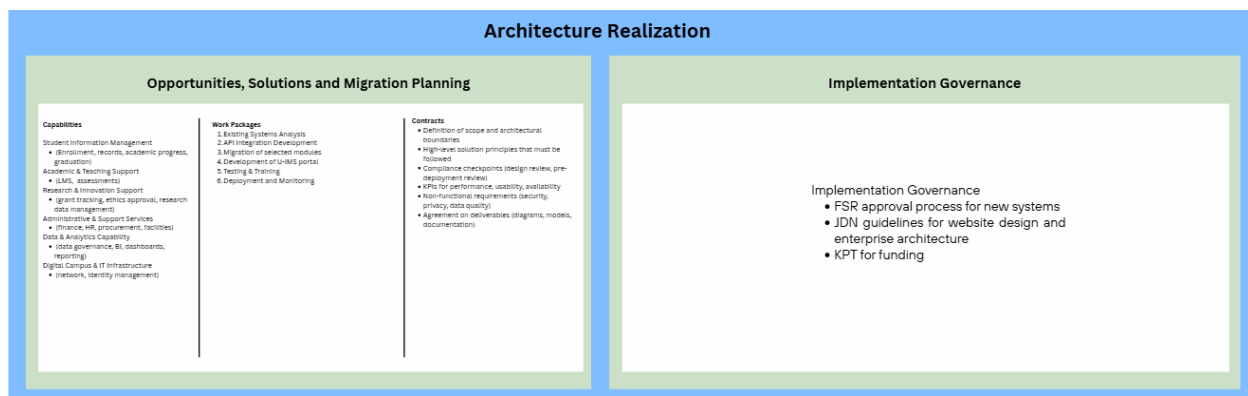


Figure 4.5 Opportunities and Solutions, Migration Planning and Implementation Governance element

Figure 4.6 shows the change management component within the EA. The Architecture Change Management (Phase H) creates a governance structure to guarantee that the university's digital systems change in an organized way rather than through random changes. It divides modifications into three categories: process simplification, gradual process incrementation, and system redesign. This guarantees that minor enhancements are implemented quickly, while high-impact changes are examined with extensive evaluation required to preserve system reliability and security.

The Architecture Review Board (ARB) manages the process, which is started by a formal change request if there are new technologies or changes in business needs. Each proposal must go through an extensive impact analysis to determine its impact on data integrity, current integrations, and the user experience. This phase ensures that all technological evolutions stay firmly connected with the university's long-term strategic strategy for 2030 by requiring official permission prior to execution.

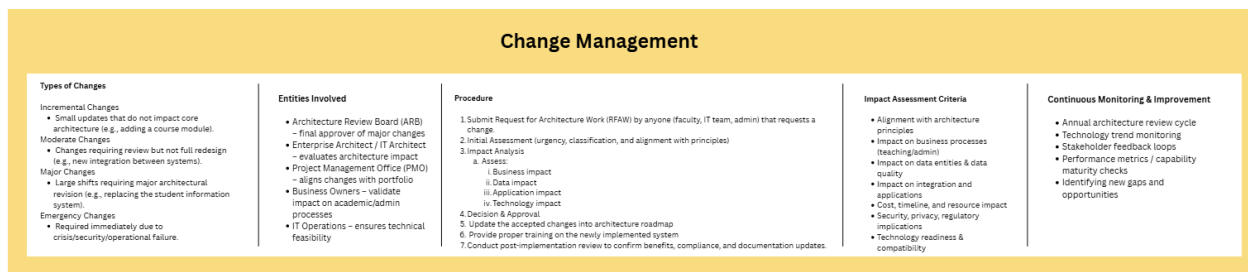


Figure 4.6 Change Management element

4.5.3 ENTERPRISE ARCHITECTURE EA – TO BE

Figure 4.7 below shows the proposed TO-BE enterprise architecture. It serves as the blueprint for UTHM on the proposed recommendations that can be improved based on the AS-IS enterprise architecture that still aligns with their business while at the same time improving efficiency. Some components have changes based on recommendations to improve business model while some principles remain the same.

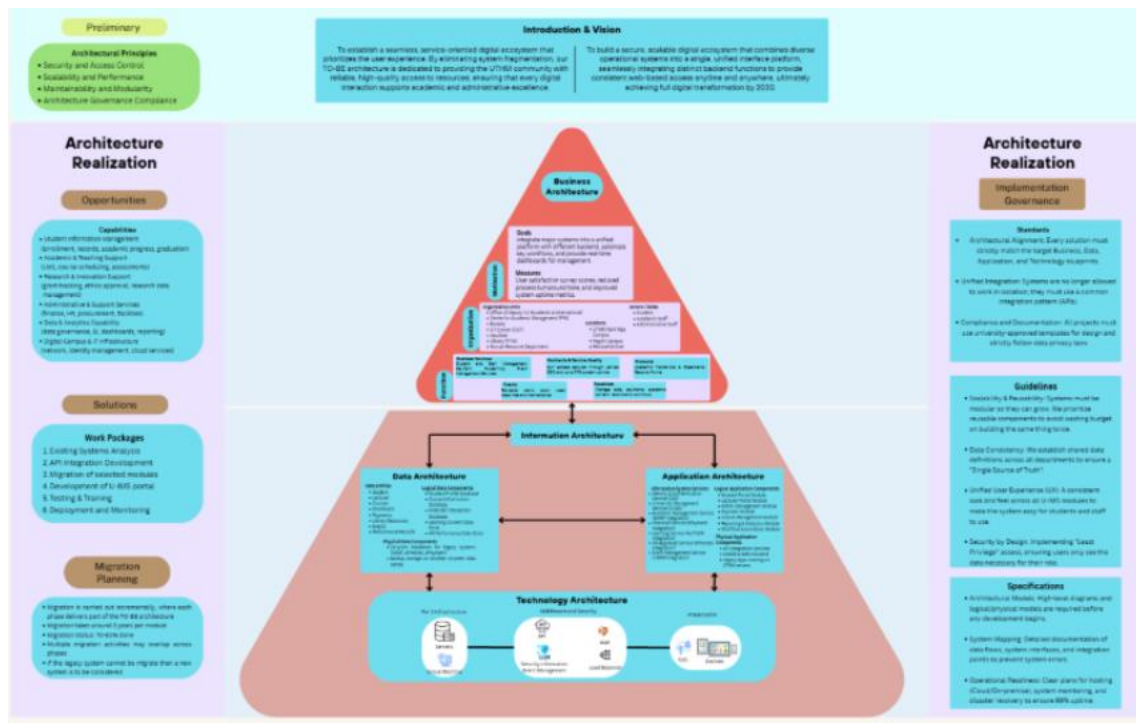


Figure 4.7 Enterprise Architecture TO-BE

4.5.4 COMPONENT EXPLANATION OF ENTERPRISE ARCHITECTURE EA – TO BE

Figure 4.8 shows the diagram that illustrating the architectural principles, visions and requirements. The primary goal of this design is to accomplish complete digital transformation by 2030. This is built on a goal to create a secure, scalable, and unified Integrated University Management System that offers all users uniform web-based access from anywhere. Several business reasons are driving this change, the most important of which is the necessity to combine distributed academic and financial data, as well as the growing demand for digital services. UTHM hopes to provide a single interface for all stakeholders, minimizing manual effort and boosting decision-making through integrated analytics.

To ensure that the system has a strong structure, the preliminary phase develops four basic architectural concepts. These serve as the "base rules" for development, stressing scalability and modularity. The principles also focus on architecture governance compliance so that all systems follow the architecture planned. Besides, security is a non-negotiable component, with a particular emphasis on data privacy, compliance, and access control, to ensure the architecture stays controlled and legally sound.

The proposed Architecture Vision (TO-BE) aims to realize UTHM's Digital Transformation 2030 agenda by establishing a unified, cloud-enabled Smart Campus ecosystem. This vision shifts the strategic focus towards high interoperability and a Single Source of Truth (SSOT), leveraging technologies like SAP Build Apps and SIEM to create a seamless, secure, and user-centric environment that integrates all academic and management functions into a single, scalable platform.

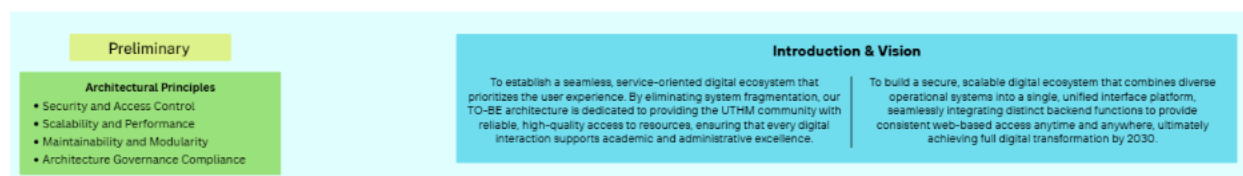


Figure 4.8 Preliminary and Architecture Vision element

Figure 4.9 below shows the TO-BE business architecture, information architecture and technology architecture.

The business architecture TO-BE remains the same as the business architecture AS-IS where it still focuses on improving the student experience and reducing the administrative workload. The architecture still involves the same organizational units and roles including academic, ICT, and administrative departments. In addition, the core function and services remain the same, such as Student Enrollment Service, Course Registration Service, Online Payment Service, and Academic Performance Service. This consistency that the existing business architecture is harmonized well with university's objective where it does not require a significant change in organizational structure or functionalities.

The TO-BE information system architecture represents a better future state than the AS-IS by moving from basic, siloed systems to an integrated, service-oriented environment. The TO-BE architecture centralizes and increases data entities such as enrollment, payments and performance records and adds standardized services like single sign-on and system integration which differ to the AS-IS architecture which primarily covers basic tasks with minimal integration. This change makes it possible for modular applications, enhanced security, improved interoperability, and increased scalability to meet academic and business requirements.

The changes from AS-IS to TO-BE technological design propose an increase in system durability, data flexibility, and continuous security monitoring. While the basic infrastructure remains focused on Tier 3 on-premises data centers in Batu Pahat and Pagoh, the TO-BE architecture incorporates offsite recovery to assure business continuity and data availability during local disasters, which is a significant improvement over the present system. Furthermore, the usage of VMware for virtualization results in a more resilient and scalable hosting environment than the present basic web application architecture. The network and security layers get major performance and monitoring improvements. The TO-BE architecture includes load balancers to manage large traffic levels. Most significantly, the security infrastructure is improved by the implementation of Security Information and Event Management (SIEM), which enables real-time threat detection and monitoring while supporting the university's current firewalls and Web Application Firewall (WAF).

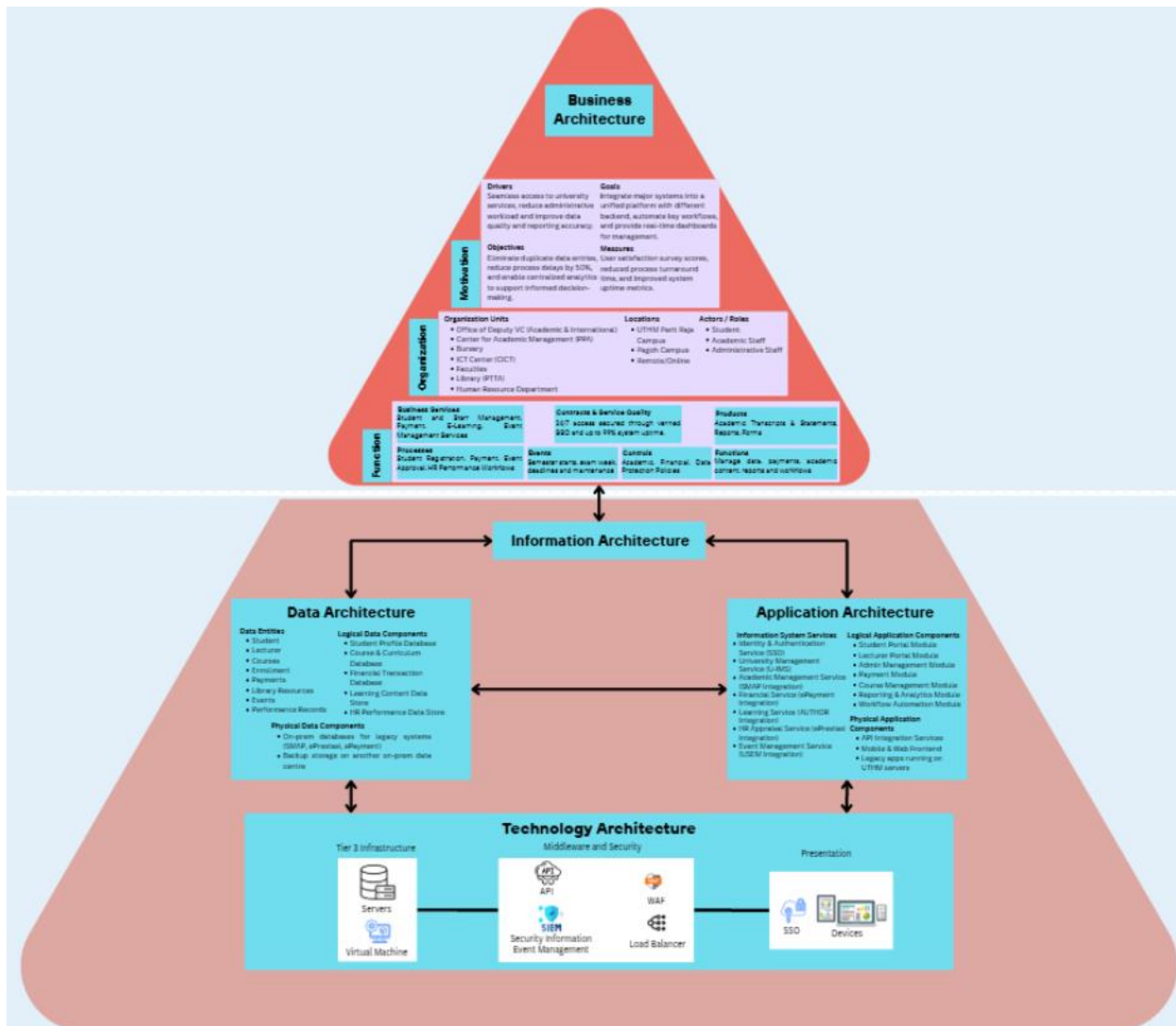


Figure 4.9 Business, Information System and Technology Architecture element

Figure 4.10 below presents the future direction of UTHM's enterprise systems. In this architecture, the focus is on modernization, integration, and digitalization. New solutions such as API integration development, and a unified U-IMS portal are introduced. These solutions aim to improve system connectivity, reduce manual work, and provide better user experience for students and staff. Data is managed more centrally, which helps improve reporting, data accuracy, and decision-making.

For migration planning, the TO-BE architecture outlines a clearer and more structured roadmap. The work packages include module migration, system development, testing, training, deployment, and monitoring. This ensures that systems changes are done step by step without disrupting daily university operations. The contracts section becomes more detailed by including compliance checkpoints, non-functional requirements such as security and privacy, and clear deliverables. Overall, the below figure shows how UTHM can move from a partially integrated environment to a fully digital, secure, and well-managed enterprise system that supports long term growth and digital transformation.

Figure 4.10 illustrates the Implementation of Governance for the university's target enterprise architecture. It is structured into a comprehensive framework consisting of three core pillars which are standards, guidelines, and specifications. The standards section establishes the mandatory rules that all new IT developments must satisfy. Architectural alignment ensures every solution fits the university's master plan, while Unified Integration mandates the use of APIs to break down system silos. Furthermore, Compliance and Documentation standards ensure all projects follow legal data privacy requirements and use standardized design templates.

Furthermore, the Guidelines provide best practices for high quality system design. Scalability and Reusability promote modular build to save costs, and Data Consistency ensures a "Single Source of Truth" across all departments. To improve accessibility, the Unified Use Experience (UX) mandates a consistent interface, while Security by Design limits data access to a "Least Privilege" basis. Additionally, the specifications act as the technical blueprints required before launching. Architectural Models provide the visual structure of the system, while System Mapping documents precise data flows and integration points to prevent errors. Finally, Operational Readiness ensures that hosting, monitoring, and disaster recovery plans are in place to support the university's 99% uptime goal.

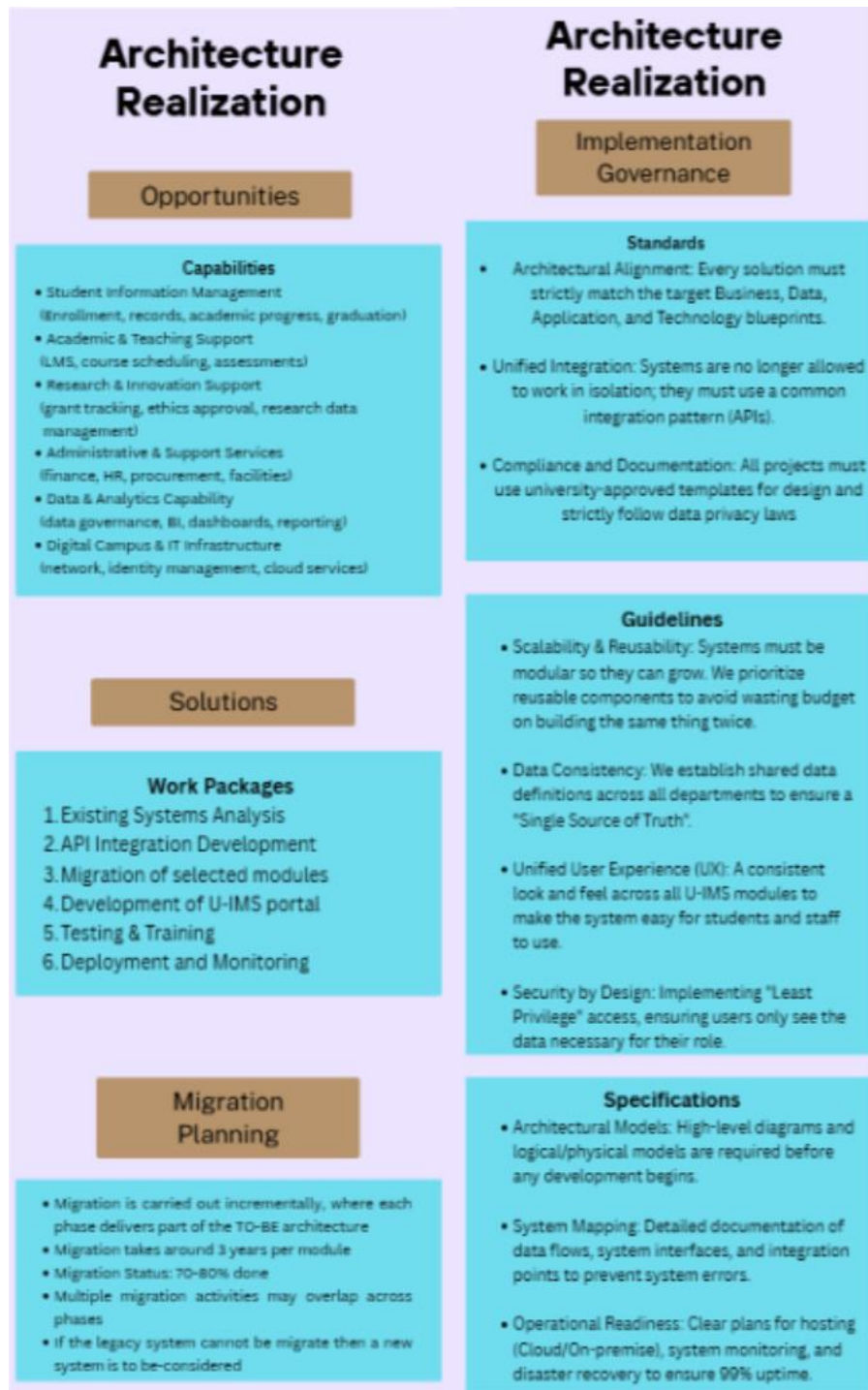


Figure 4.10 Opportunities, Solutions and Migration Planning and Implementation Governance
Architecture element

4.5.5 SYSTEM ARCHITECTURE

Figure 4.11 shows the cloud system architecture designed for the academic management platform. As the applications are developed using cloud platforms called SAP Build Apps and SAP Build Process Automation, a cloud-based architecture is used.



Figure 4.11 Cloud System Architecture for the System

4.5.6 COMPONENT OF SYSTEM ARCHITECTURE

The three user's role which are Student, Lecturer, and the Admin can access the applications through their web browsers and mobile devices. Students use the system for registering for courses and viewing student profile. Lecturers access the system for viewing student lists and approving student course registration. Admins use the system for managing course and section and allocating time schedules and location for courses. This layer communicates with the backend through the Internet.

The Application layer shows the Services provided in the system. The services including registering courses and the approval process, viewing profile, and managing courses and sections details, class schedules, and class location.

The system uses NoSQL and in-memory database type. The main data component used are the app variable, page variable, and the on-device storage database. The app and page variables are temporary data stored when the application is running, while on-device storage database is a database that allows some data to be stored locally on user's device when the application offline.

4.6 PROJECT DESIGN

The “View Student List for Each Section” subsystem of the Course Registration System is described in detail in this section. The designed is focuses on the easy access and retrieval of student enrollment data by lecturers and administrators. In this section, UML diagrams are done to show the interaction flow and system requirements. The use case diagram defines the the functional scope of the system while the use case description provides a detailed explanation of the workflow. The sequence diagram shows the interaction between actors and system components during execution while the activity diagram shows the process flow of user actions and system responses.

4.6.1 USE CASE DIAGRAM

Figure 4.12 shows a use case diagram for “View Student List for Each Section”. It includes two users who are Lecturers and Admins. The use case shows that both users can select a course, choose a section and view the student list. The admin also allows to manage the student list. The diagram explains how each user interacts with the systems.

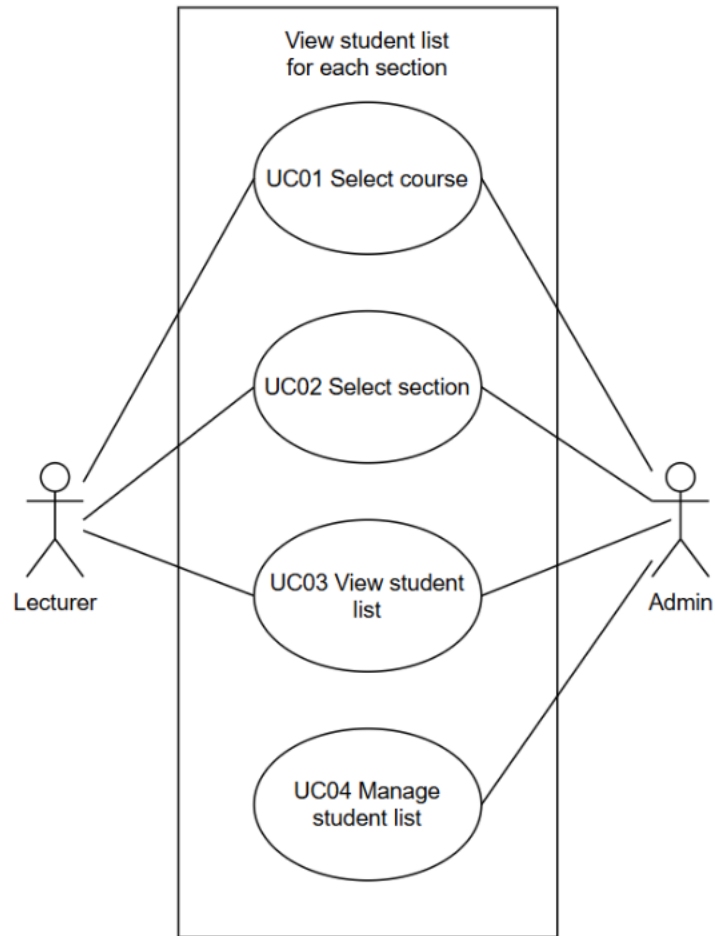


Figure 4.12 Use Case Diagram

4.6.2 USE CASE DESCRIPTION

Table 4.1 shows a use case description about how lecturers or admins view the student list for a specific course section. The user must first log into the system then choose the option to view the student list, select course and then select section. The system will show the list of students enrolled in that specific section. If there are no students registered yet, the system will display a message of “No students enrolled”.

Table 4.1 Use Case Description

Use Case: View Student List for Each Section
ID: UC001
User Story Description As a lecturer or admin, I want to view the student list for a specific course section, So that I can monitor student enrollment and manage academic records.
Flow of events: 1. The lecturer or admin logs into the system by accessing the login page and entering valid login credentials. 2. The lecturer or admin selects the “View student list” option. 3. The system displays lists of available courses. 4. The lecturer or admin select the course from the list. 5. The system displays the available sections under the selected courses. 6. The lecturer or admin selects section for the course. 7. The system retrieves and displays the student list for the selected section.
Alternative flow <i>n</i>: n.1: If the selected section has no enrolled students, the system displays a “No students enrolled” message.
Acceptance Criteria Postcondition: 1. The student list for the selected section is successfully displayed.

Precondition:

1. The lecturer or admin must be logged into the system.
2. The selected course and section must exist in the system.
3. Students' enrollment must exist for the section.

Other conditions:

1. Only users with authorized roles can access this function.

Exception flow:

1. If the system detects an error, an error message is displayed and the lecturer or admin is unable to proceed.

4.6.3 ACTIVITY DIAGRAM

The activity diagram in Figure 4.13 shows the action flow when a lecturer or admin views the student list for each section. The process starts with the user log in and the system check the credentials whether the login is valid or not. If the login is invalid, an error message displayed but if it is valid, the dashboard will be displayed. The user can then select the option to view the student list by choosing the course and sections and the system will check whether students are enrolled. The system will display either the student list or an error message saying there are no students enrolled and the process ends.

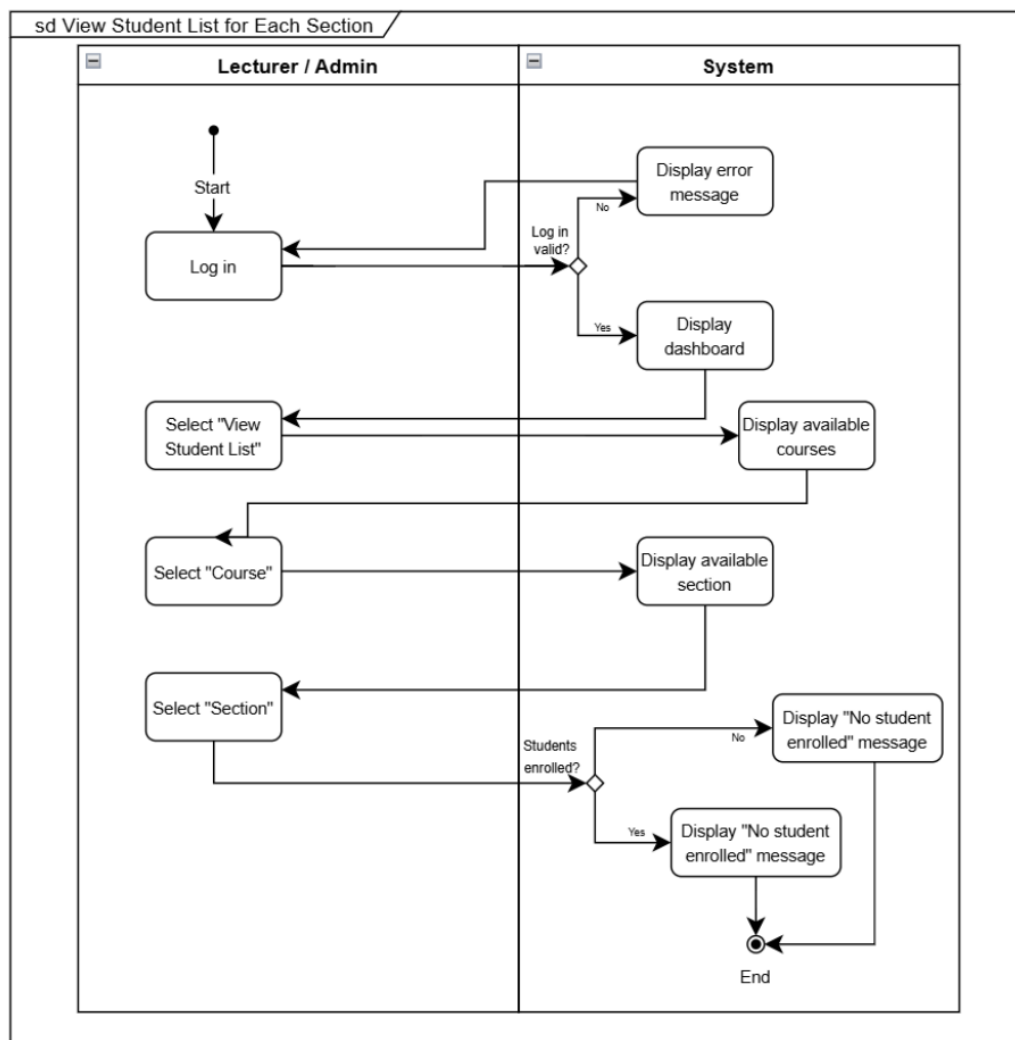


Figure 4.13 Activity Diagram

4.6.4 SEQUENCE DIAGRAM

Figure 4.14 is the sequence diagram that shows an interaction between the lecturer or admin and the system while view the student list for each section. It explains on how the user need to log in first then selects course and section, then how the system responds by displaying the student list or message if there are no students enrolled in specific section of a course.

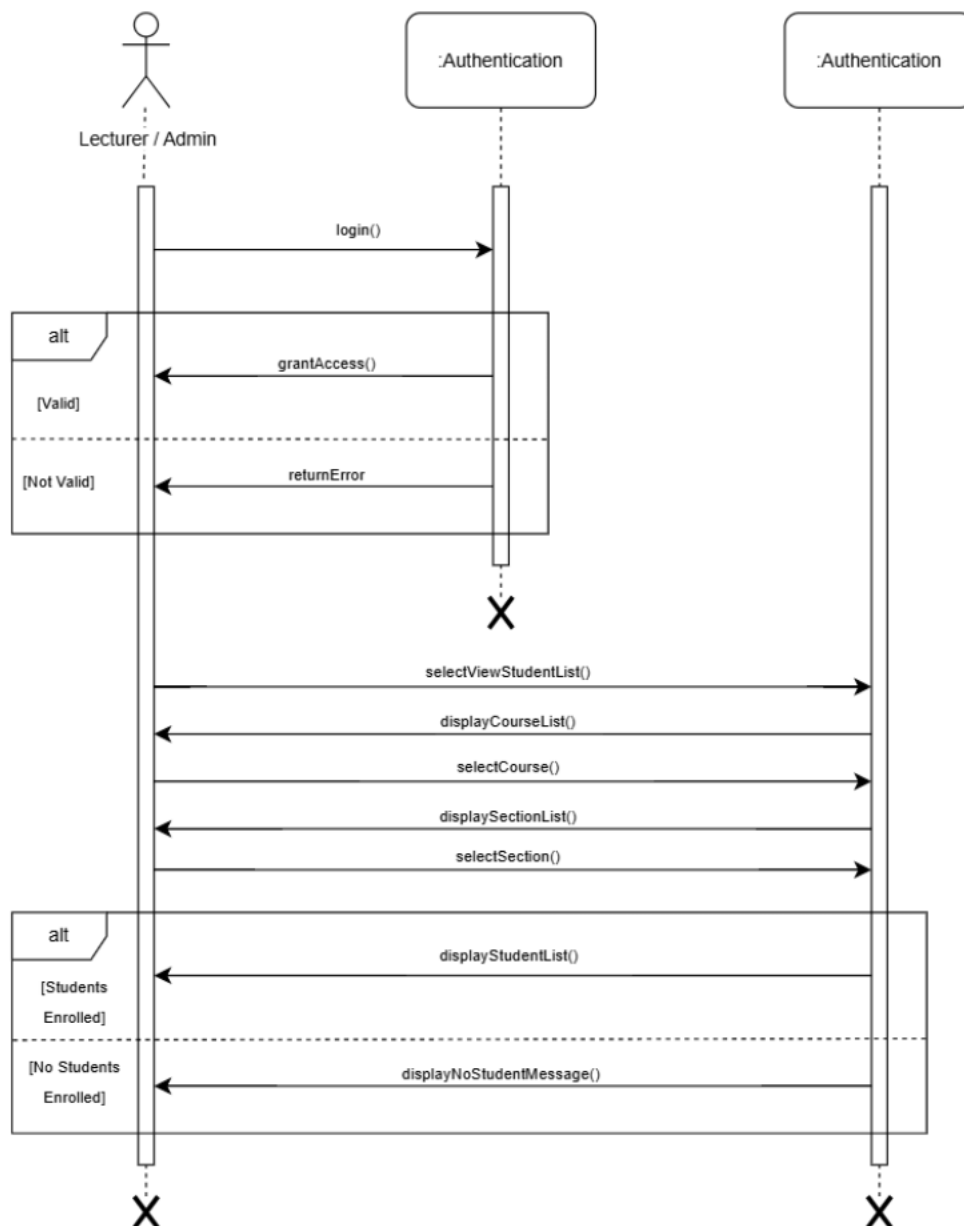


Figure 4.14 Sequence Diagram

4.7 DATABASE DESIGN

Figure 4.15 is an Entity Relationship Diagram (ERD) that represents the data structure for “View Student List for Each Section” subsystem. A user who are lecturer or admin can view multiple courses at a time and each course contains many enrollments. This enrollments represent the student who registered in that section of the course. Student details stored in Student entity to indicate which students enrolled in specific section.

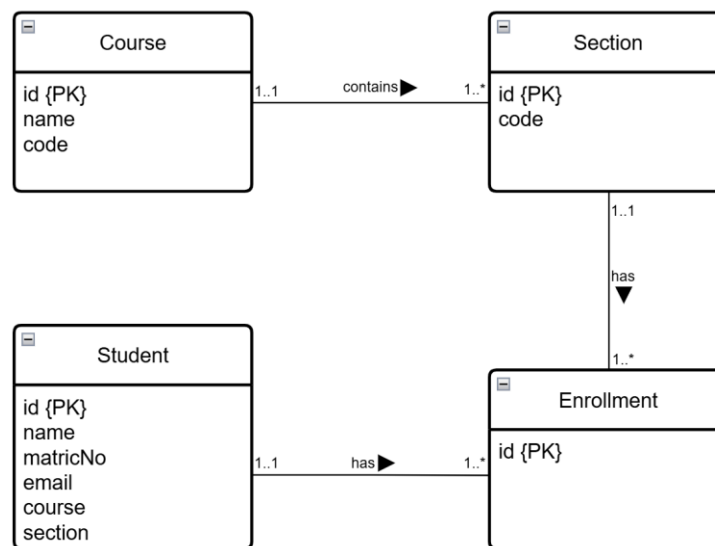


Figure 4.15 Entity Relationship Diagram

4.8 INTERFACE DESIGN

Figure 4.16 shows the interface that allows the user to select a course from the course list provided. After choosing the course and list, the user proceeds to the next step by clicking the “Next” button. The “Next” button will guide to the student list page while the “Back” button returns the user to the previous page. For Figure 4.18, it shows the page of students list registered under the selected section. The user can return to the home page using the “Return to Home Page” button. Moreover, the interface allows the user to register new students by clicking on the “Add Student” button and selecting “Details” to view student details.

**UTHM**
Universiti Tun Hussein Onn Malaysia

UTHM COURSE LIST

COURSES

SECP3744 ENTERPRISE SYSTEMS DESIGN AND MODELLING (WBL) ▾

SECTIONS

SECTION 03 ▾

NEXT

Figure 4.16 Select Course and Section Page

**UTHM**
Universiti Tun Hussein Onn Malaysia

SECP3744 ENTERPRISE SYSTEMS DESIGN AND MODELLING (WBL)

SECTION 03

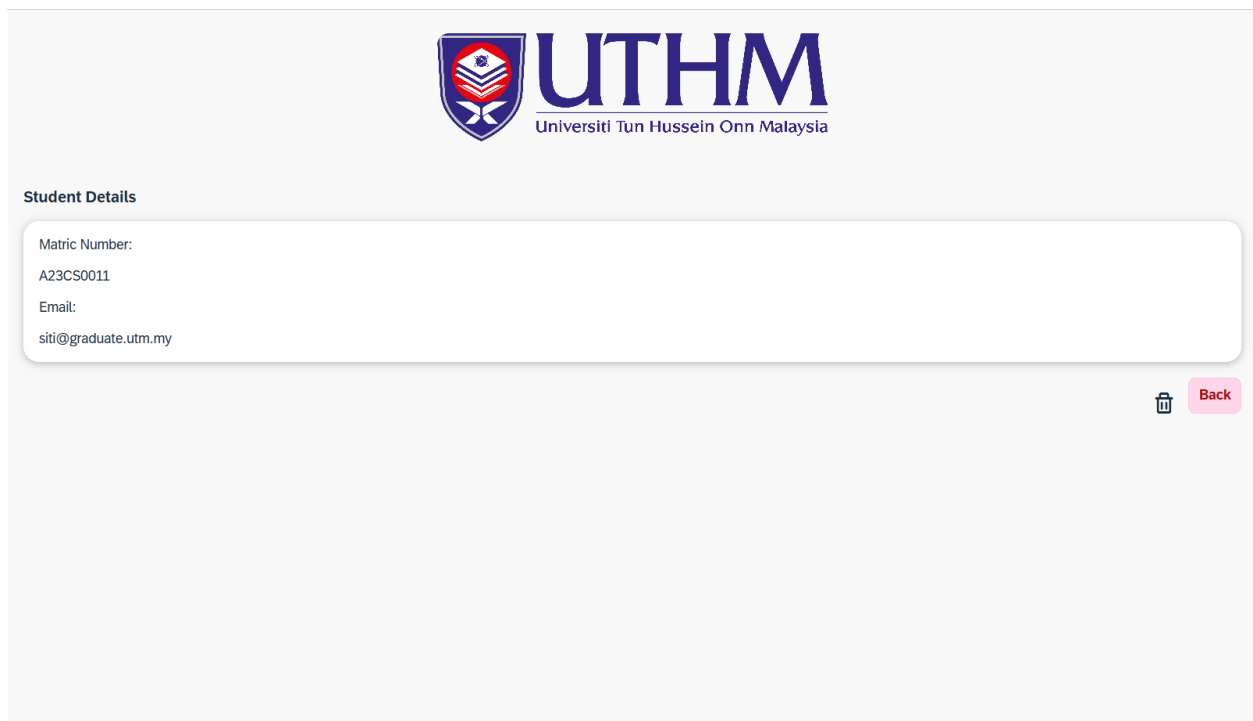
Add Student

RETURN TO HOME PAGE

Student Name List	
Siti bin Aminah	Details >
Faiz bin Hasan	Details >
Aishah binti Zack	Details >
Khairut bin Mohd	Details >
Zul bin Razak	Details >
Ali bin Abu	Details >

Figure 4.17 Student List Page

Figure 4.18 shows the Student Details page where it shows the student matric number and email. It provides the user with a “Back” button to go to the previous page and the trash icon to delete the entry of the student. Figure 4.19 shows the “Add Student” form which is the input interface for registering a new student for specific sections of the course. The newly registered student will be displayed on the “Student List” page in Figure 4.17.



UTHM
Universiti Tun Hussein Onn Malaysia

Student Details

Matric Number:
A23CS0011

Email:
siti@graduate.utm.my

 [Back](#)

Figure 4.18 Student Details Page

name

course

section

matricNo

email

Figure 4.19 Add Student Page

4.9 SUMMARY

This chapter has detailed the comprehensive analysis and design of the proposed Enterprise Architecture for Universiti Tun Hussein Onn (UTHM). The study began with an analysis of UTHM's organizational structure and current system environment, identifying critical challenges such as data silos, manual data transfer, and reliance on legacy infrastructure like SMAP and ePayment. The requirements gathering phase, conducted through interviews and site visits, highlighted the need for a unified platform to support the university's 20230 digital transformation vision.

Based on these findings, the chapter detailed the “AS-IS” architecture to document the existing gaps, followed by the proposed “TO-BE” architecture. The proposed design adopts the TOGAP framework to restructure the Business, Data, Application, and Technology layers, ensuring better alignment between IT capabilities and UTHM’s strategic objectives. Key improvements include the introduction of a hybrid database model (combining SQL and NoSQL) to handle diverse academic data and the implementation of a robust Implementation Governance framework to standardize future IT developments. Ultimately, this chapter establishes the architectural blueprint required to transition UTHM towards a fully integrated, cloud-enabled Smart Campus environment.

CHAPTER 5

ANALYSIS AND DESIGN

5.1 INTRODUCTION

This chapter explains how the Student Course Registration subsystem was implemented specifically on the View Student List for Each Section module. The chapter describes the Waterfall technique used to construct the subsystem which covers the steps of requirement analysis, design, development and testing. Furthermore, the database creation process, the system flow and the functional modules developed with SAP Build Apps are also covered. This chapter describes how the specifications given in the previous chapter were used for converting the designed system into a functional prototype.

5.2 SYSTEM DEVELOPMENT

The system was developed using the Waterfall methodology in which was chosen because its methodical and systematic approach. For the development process must be clear and consistent, each phase had to be finished before going on to the next.

Firstly, the requirement phase was carried out using data obtained from the UTHM industrial visit and interviews. Functional requirements for the Student Course Registration subsystem were identified during this phase.

Next, the design phase then included the development as described in Chapter 4 which includes the use case diagrams and description, activity diagrams, sequence diagrams, database design and interface design. These designs functioned as a framework for the system's execution. The core data entities including Student, Course, Section and Enrollment were used in the Entity Relationship Diagram (ERD).

Moreover, the system was implemented using SAP Build Apps which is a cloud-based platform during the development stage. The development of user interface screens allowed for the selection of courses and sections. Enrollment and students' data was filtered by logic flows according to the chosen course and section. Furthermore, student enrollment records were retrieved and temporarily stored using data variables. The goal of this step was to make sure the subsystem operated in line with the intended workflow.

During the testing step, the subsystem was tested to ensure that each chosen courses and sections presented the right student list. Unit testing like section filtering and course selection was first evaluated before testing the entire workflow. This step involved fixing any mistakes that were found. The subsystem is developed as working prototype in SAP Build Apps preview function.

5.3 CREATE DATABASE

The “View Student List for Each Section” subsystem makes use of in-memory database technologies from SAP Build Apps as shown in Figure 5.1 until 5.5. While on-device storage is used to save specific data when the system is offline, app variables and page variables are used to temporarily stored data while the system is operating. This method offers quick data retrieval and enhances system efficiency.

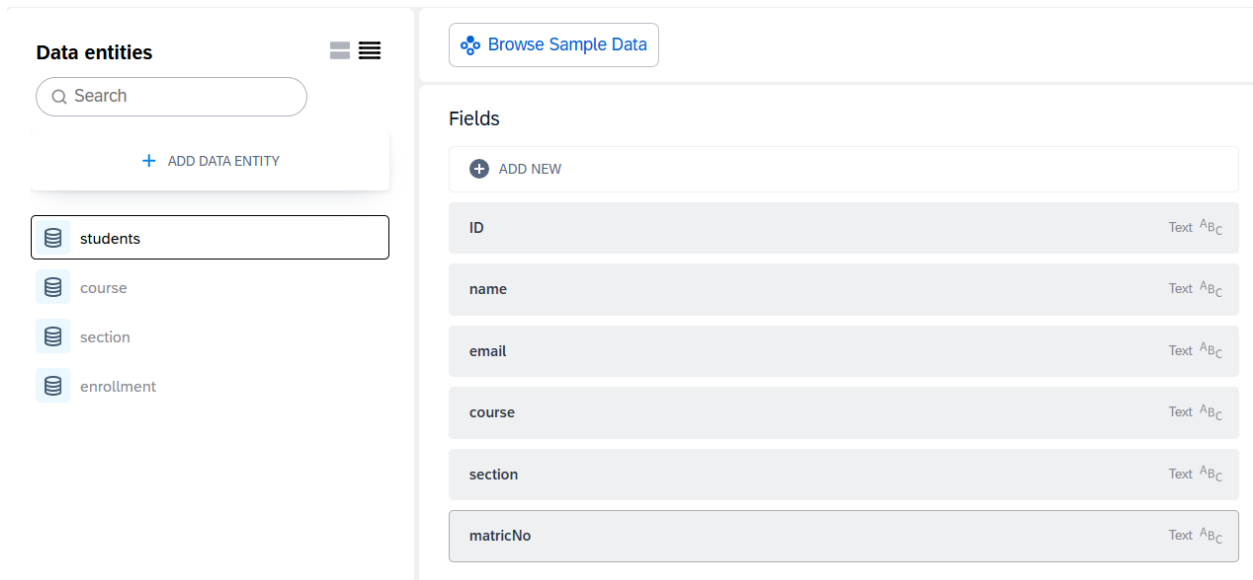


Figure 5.1 students database

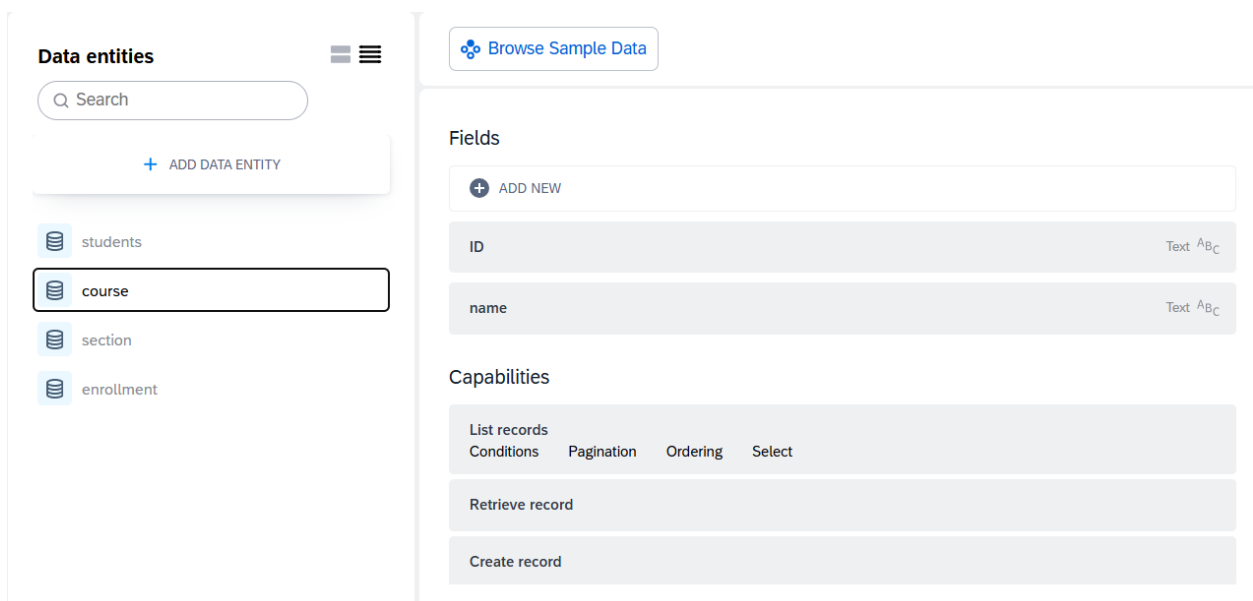


Figure 5.2 course database

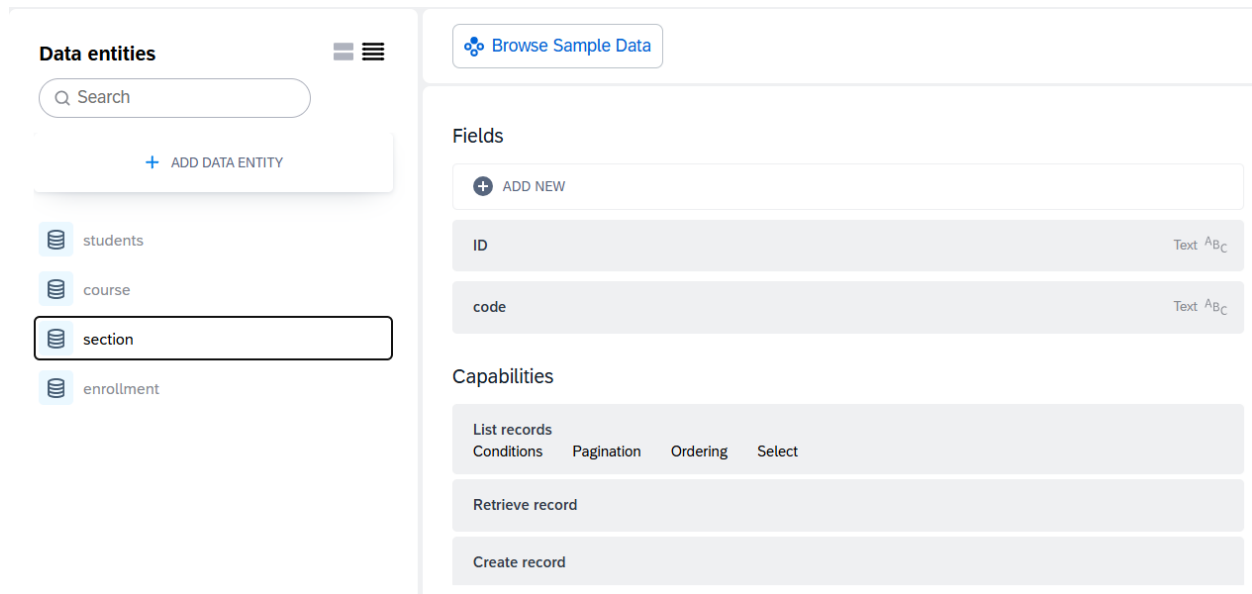


Figure 5.3 section database

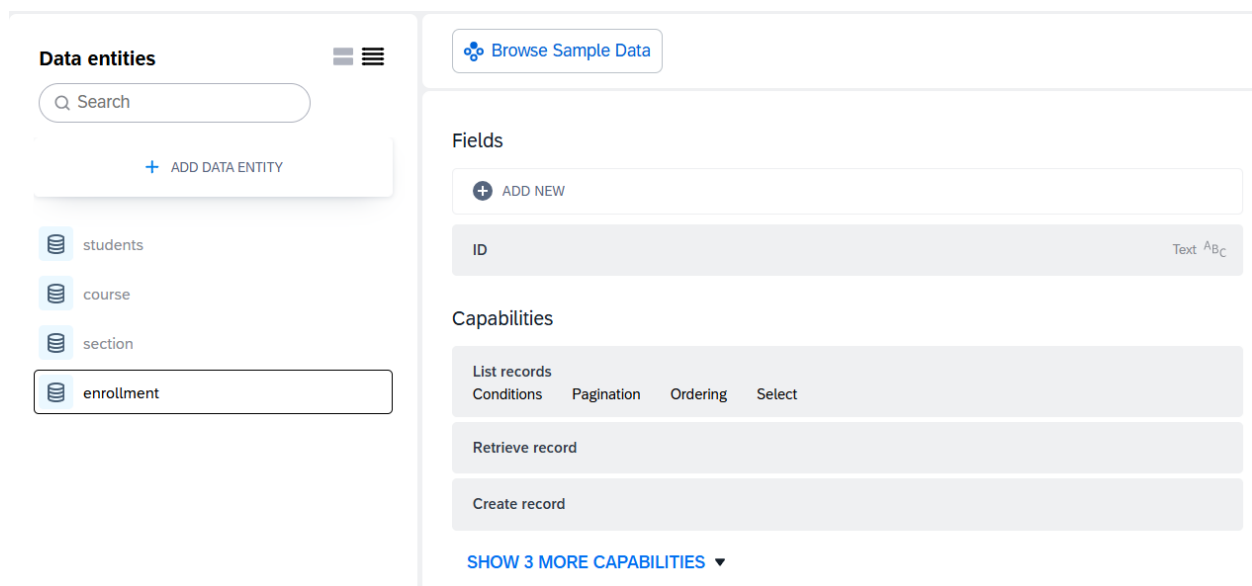


Figure 5.4 enrollment database

5.4 CODING OF SYSTEM PROCESS

This subtopic focuses on implementing the system's key functional modules of the system specifically on the "View Student List for Each Sections" subsystem. There are three main parts of the implementation including the input, process and output.

5.4.1 Input Module

Through this input module, admin and teachers can interact with the system to view student enrollment information. From the list of available courses, users choose a course and then a particular section. Furthermore, the provided forms can be used by admin to register new students to the course's section. These inputs are recorded using form elements from SAP Build Apps.

5.4.2 Process Module

The "View Student List for Each Section" module's core logic is managed by the process module. Following the selection of a course and section, the system filters enrollment data according to the chosen section. The related student enrollment information is then retrieved by the system from the database.

5.4.3 Output Module

The output module shows the processed data's outcomes. The system displays a full listing of all students registered in the chosen course section. If there are no students registered for the chosen section, the system will show nothing. Moreover, the output interface allows users to see specific student information.

5.5 SUMMARY

In conclusion, the implementation of the Student Course Registration subsystem specifically the View Student List for Each Section module has been covered in this chapter. The Waterfall technique was used throughout the development process to guarantee a methodical and organized execution. Moreover, SAP Build Apps were used in the subsystem's development and in-memory database technologies were used to enable effective data handling. Besides, student enrollment data was successfully displayed using the input, process and output modules based on the chosen course and section. Lastly, this chapter explains how a functioning subsystem was created from the system design.

CHAPTER 6

CONCLUSION

6.1 INTRODUCTION

This chapter provides a comprehensive review of the developed system, focusing on the evaluation of its functional performance, and its overall impact on student course registration. The specific roles played by each module in streamlining student course registration and administrative are examined to demonstrate the project's practical utility. Additionally, the technical constraints identified during the implementation phase are documented to provide a realistic assessment of the system's current capabilities within the SAP Build environment. The chapter concludes with strategic suggestions for future enhancements aimed at improving system integration and automation.

6.2 SYSTEM CONTRIBUTION

This project consists of seven modules, which are Student Course Registration module, Manage Course Credit Hour module, Section Scheduling module, Admin Dashboard module, Manage Student Profile module, Course Approval module, and View Student List for each Section module.

The Student Course Registration module is used to allow courses and sections to be selected and registered online. Besides, courses and sections can be deleted, and a real-time list of registered courses and sections can be viewed. The errors are minimized because real-time registered courses and sections can be viewed and confirmed before the course registration is submitted.

The Section Scheduling module is used to manage the venue and class time of each section. It is ensured that no sections clash with the class time or venue of other sections. The classroom usage is optimized and schedule clashes due to human error are prevented.

The Course Management module is used to manage course information. In this module, course can be added and managed by admins. Course management includes changes to the course code, course name, and credit hour. Moreover, courses can be deleted too.

The Admin Dashboard module is used to allow administrators to monitor and manage the course registration system. Registration statistics and analytics can be viewed on the main Dashboard page. For pending registrations, administrators can view the list of students awaiting Academic Advisor approval filtered by course. System settings such as registration period, credit limits, and notification preferences can be configured through the Settings page. Activity Logs are available to track all actions performed in the system.

The View Student List for Each Course module is used to allow instructors and administrators to view a full list of every student registered in a certain course section. Student enrolment can be monitored more efficiently by lecturers and administrators. Moreover, manual record keeping is reduced, and enrolment data accuracy is increased.

The Course Approval module is used to allow registered courses to be reviewed and approved by academic advisers. It is ensured that registration data is transmitted accurately from the student to the academic advisor without manual handling. By guaranteeing that every registration is formally evaluated through a standardized workflow, academic control is maintained, and communication errors are minimized.

The Student Profile module is used to collect all student data including personal and academic information into one subsystem. Moreover, access to student data is supported for administrative purposes when needed. Student monitoring is improved, errors are minimized, and the redundant data entry is reduced.

6.3 SYSTEM CONSTRAINT

While the proposed architecture introduces significant operational improvements, several technical constraints were identified within the current design phase. A primary technical limitation is the system's dependency on stable internet connectivity. Since the solution is architected using the cloud-native capabilities of SAP Build Apps and remote NoSQL data storage, continuous network access is required for functionality. Consequently, latency or temporary inaccessibility to real-time scheduling data may be experienced in environments with limited bandwidth.

Furthermore, the integration with the university's existing legacy infrastructure presents a notable challenge. The transition from traditional relational databases –such as those used in SMAP and ePayment –to a modern NoSQL architecture necessity of complex data synchronization. Currently, this interoperability is proposed to be achieved through API bridges, which may impose limitations regarding data transfer speeds and format compatibility during the interim migration period.

Finally, regarding the development environment, the use of a Low-Code Development Platform (LCDP) was selected to facilitate rapid prototyping and proposal visualization. While this approach effectively demonstrates the system's core value and feasibility, the implementation of highly complex, customer algorithms –such as AI-driven automated conflict resolution –is bounded by the pre-defined flow functions available within the platform.

6.4 FUTURE SUGGESTION

The project has considerable amounts of opportunity to grow in the future, going from a number of individual modules to an integrated, professional system. The complete integration of the seven separate elements into a single ecosystem is the main plan for future development. Data input in the "Manage Student Profile" module does not now automatically appear in the "Student Course Registration" or "View Student List" modules due to the modules' current isolation. Through the use of SAP Business Technology Platform (BTP) to develop a centralized data backend, the system may provide a single "source of truth." This would eliminate data duplication and the possibility of human input mistakes by ensuring that every change is made in one module. For example, a student changing their credit hours is reflected across the whole system in real time.

The next suggestion is the implementation of process simplification and workflow automation. The system can be configured to have automated processes using technologies like SAP Build Process Automation. For instance, a submission in the "Student Course Registration" module could instantly alert a lecturer in the "Course Approval" module. This greatly reduces the workload for administrators by establishing a proactive system as opposed to a reactive one. Moreover, security requires the implementation of Role-Based Access Control (RBAC). This would guarantee that just registration and profile tools are visible to students, and administration modules such as "Section Scheduling" and "Section Allocation" are only accessible by those with authorization.

6.5 REFLECTION

This assignment helped me in gaining better understanding of enterprise systems and Enterprise Architecture (EA). The use of TOGAF framework in this project allowed me to understand the concept of EA in more structured and organized. By analyzing the requirements for UTHM, the importance of properly planning for a project is clearly realized. Although there are difficulties faced during the project, these challenges were overcome through discussion through team members and guidance from lecturer.

Moreover, this project contributed to improve skills like time management, teamwork and documentation. Overall, the assignment was beneficial as it provided practical knowledge and offer valuable preparation to the real enterprise systems and EA field.

6.6 SUMMARY

In summary, the development of the automated course registration system successfully demonstrates the feasibility of using low-code technology to improve course registration workflows. While the system provides significant benefits in terms of data accuracy and communication transparency, it remains subject to technical dependencies such as internet connectivity and platform specific logic boundaries. The identified constraints and future suggestions provide a roadmap for transitioning this prototype into a more comprehensive, centralized ecosystem. Ultimately, the project fulfills its primary objective of modernizing the course registration process through automated approvals and secure data management.

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